ENT ATLAS

IMAGING

Imaging conditions

Key facts and case images for the core conditions, with the advanced list

· November 2026

IMAGING

Core conditions

The common conditions every resident should recognise, each with its key imaging findings, top differentials, a short checklist and representative scans.

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

Malignant External Otitis or Necrotizing External Otitis

IMAGING FINDINGS

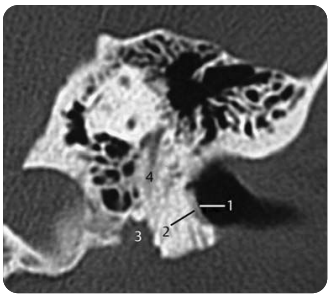
- Soft tissue in the external canal with erosion of its floor and the tympanic bone
- Spread below the skull base: loss of fat at the stylomastoid foramen, parapharyngeal and masticator spaces, temporomandibular joint
- Later: skull base osteomyelitis of the clivus, jugular foramen and petrous apex; venous thrombosis

TOP DIFFERENTIALS

- Carcinoma of the external canal
- Canal cholesteatoma
- Keratosis obturans

CHECKLIST

- Facial nerve and lower cranial nerves
- Sigmoid sinus and jugular vein; follow-up MRI changes lag behind clinical cure



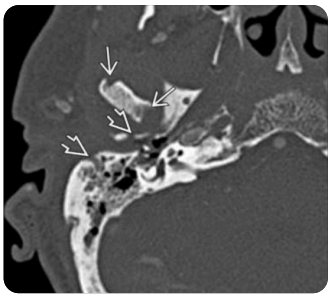
CT, coronal. Necrotizing lesion in the floor of the external auditory canal (1). Clinically, erosive bone lesions (2) with purulent debris and denuded bone…



Axial bone CT shows EAC opacification with focal anterior wall and floor of middle ear erosion in this diabetic patient with painful otorrhea and earl…



Coronal bone CT in the same patient demonstrates anterior EAC wall bony destruction accompanied by complete opacification of the EAC. The middle…



Axial bone CT reveals EAC opacification associated with multiple areas of erosive bony change. The mandibular condyle is also eroded,…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1010.

MIDDLE EAR & MASTOID · TRAUMA

Luxation of the Ossicular Chain

IMAGING FINDINGS

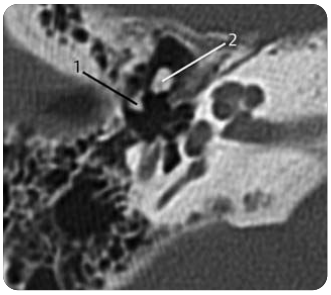
- Separation of the incudostapedial joint; incus rotated or displaced
- Loss of the "ice-cream cone" between malleus head and incus body
- Associated temporal bone fracture through the canal and middle ear

TOP DIFFERENTIALS

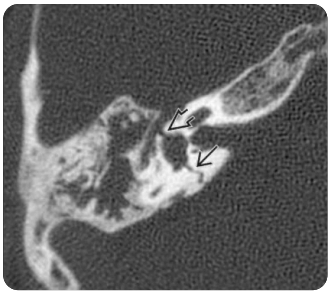
- Ossicular erosion from chronic otitis
- Congenital ossicular malformation
- Partial volume on a single slice

CHECKLIST

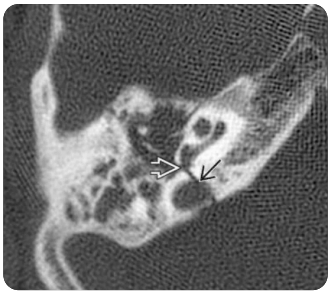
- Compare with the other ear on the same slice
- Check facial canal, tegmen and otic capsule



CT, axial. Traumatic luxation with dislocation of the connection between incus (1) and malleus (2) is seen. The typical icecone configuration of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…



Axial bone CT more inferiorly in the same patient shows typical inferior extension of the fracture across the jugular foramen, into the round windo…



Axial bone CT shows a chronic transverse fracture line extending through the otic capsule to the basal turn of the cochlea, where abnormal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

MIDDLE EAR & MASTOID · BONE DISEASE

Otosclerosis, Radiologic Signs around the Oval Niche

IMAGING FINDINGS

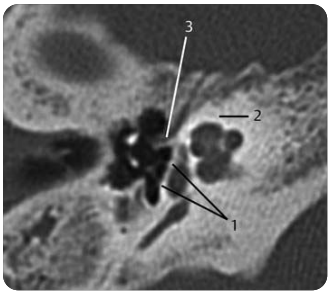
- Lucent focus at the fissula ante fenestram, anterior to the oval window
- Thickened footplate or obliterated oval window niche
- Cochlear form: pericochlear lucent halo ("double ring")

TOP DIFFERENTIALS

- Tympanosclerosis
- Osteogenesis imperfecta
- Paget disease of the temporal bone

CHECKLIST

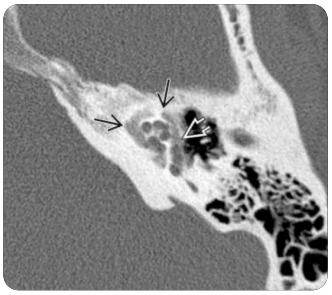
- Compare both ears: usually bilateral
- Report round window involvement and cochlear disease before stapes surgery



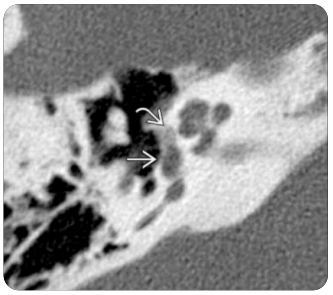
CT, axial. At the edges of the footplate, two densities are visible (1), indicating osseous thickening, which may be considered as otosclerotic foci,…



Coronal right temporal bone CT shows a lytic focus anterior to the oval window, the typical appearance and location of an otospongiotic plaque o…



Axial left temporal bone CT shows cochlear otosclerosis as osteolytic foci surrounding the cochlea. Concurrent fenestral otosclerosis is noted as bon…



Magnified axial right temporal bone CT in a young adult with progressive conductive hearing loss demonstrates typical otospongiotic plaque of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1120.

MIDDLE EAR & MASTOID · TUMOURS

Glomus Tympanicum (Paraganglioma)

IMAGING FINDINGS

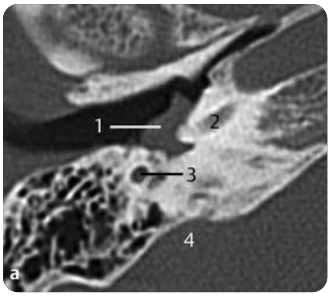
- Small enhancing mass on the cochlear promontory
- Floor of the middle ear intact
- Flow voids only in larger tumours

TOP DIFFERENTIALS

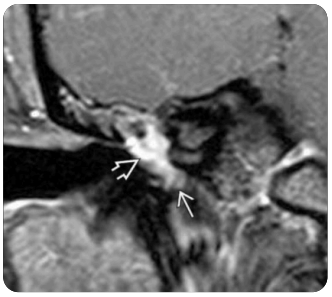
- Glomus jugulare (floor eroded, from the jugular foramen)
- Aberrant internal carotid artery
- Dehiscent jugular bulb

CHECKLIST

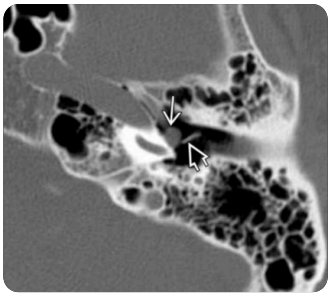
- Check the hypotympanic floor and jugular bulb
- Ossicles and extension into the mastoid



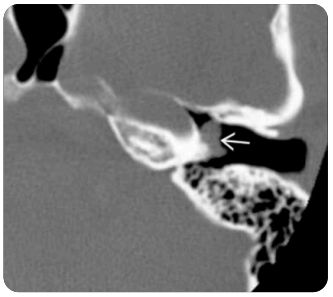
CT, axial. On this axial slice at the level of the hypotympanum, a smooth-bordered lesion (1) is seen, without any destruction of the cochlea (2) or…



Coronal T1 C+ FS MR demonstrates a large GTP filling the middle ear cavity. The floor is intact, separating the tumor from the jugular bulb below.…



An axial bone CT reveals an ovoid glomus tympanicum tumor on the low cochlear promontory abutting the manubrium of the malleus. The…



Axial left temporal bone CT shows a multilobular soft tissue mass on the low cochlear promontory, consistent with a diagnosis of GTP.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1064.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Cholesteatoma of the Middle Ear

IMAGING FINDINGS

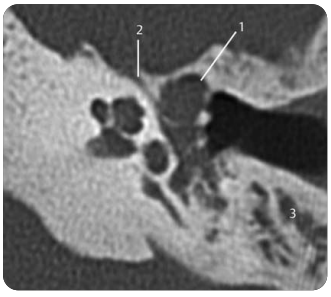
- Soft tissue in Prussak space (attic) or the posterior mesotympanum
- Blunted scutum; eroded, medially displaced ossicles
- Bright on non-EPI diffusion MRI; no enhancement

TOP DIFFERENTIALS

- Granulation tissue or effusion (no bone erosion)
- Cholesterol granuloma (bright on T1)
- Glomus tympanicum (enhancing, on the promontory)

CHECKLIST

- Five questions: tegmen, lateral canal fistula, facial canal, sinus tympani, ossicles
- Non-EPI DWI for residual or recurrent disease



CT, axial. The most frequently encountered mass in the middle ear is cholesteatoma, and therefore it is (also) shown in this chapter. In this…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…



Coronal bone CT shows a large pars flaccida cholesteatoma as a soft tissue mass within the right middle ear and mastoid cavity. There is fistulatio…



Axial bone CT reveals a large cholesteatoma in the middle ear cavity with scalloping of the walls. There is erosion of the anterior portion of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

For personal study. Follow local protocols and senior advice for patients.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Labyrinthine Fistula due to Cholesteatoma

IMAGING FINDINGS

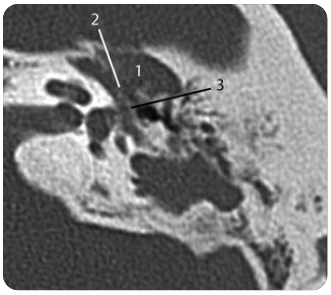
- Loss of the bony cover of the lateral semicircular canal next to cholesteatoma
- Best seen on coronal and Pöschl or Stenvers reformats
- Labyrinthine fluid on T2 may be reduced if labyrinthitis follows

TOP DIFFERENTIALS

- Partial volume of thin normal bone
- Superior canal dehiscence
- Post-surgical defect

CHECKLIST

- Report fistula site and size before surgery
- Confirm on thin slices in two planes



CT, axial. Note the smooth-bordered opacification suggestive of cholesteatoma in the anterior part of the middle ear (1), at the level of the…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…



Coronal bone CT shows a large pars flaccida cholesteatoma as a soft tissue mass within the right middle ear and mastoid cavity. There is fistulatio…



Axial bone CT reveals a large cholesteatoma in the middle ear cavity with scalloping of the walls. There is erosion of the anterior portion of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

MIDDLE EAR & MASTOID · CONGENITAL

Congenital Cholesteatoma

IMAGING FINDINGS

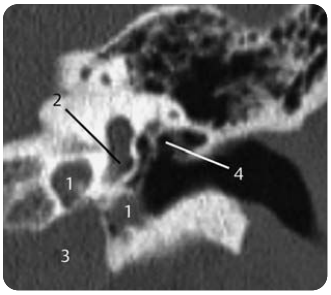
- Round, well-defined pearl behind an intact drum, usually anterosuperior mesotympanum
- Ossicles displaced or eroded, especially the incus long process and stapes
- Bright on non-EPI DWI; no enhancement

TOP DIFFERENTIALS

- Acquired cholesteatoma
- Glomus tympanicum
- Facial nerve schwannoma of the tympanic segment

CHECKLIST

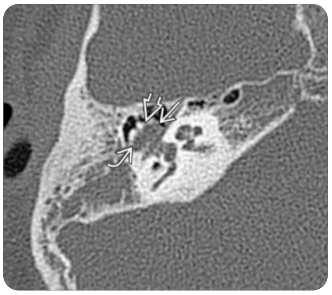
- Intact drum and no history of otitis
- Look for mastoid and petrous apex extension



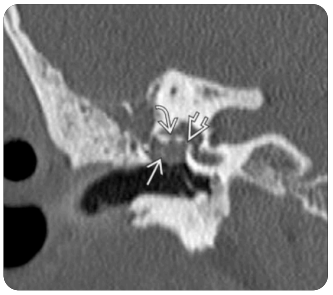
CT, coronal. CT excluded deeper invagination, but revealed other pathology. Coincidentally, an expansive infralabyrinthine…



Coronal temporal bone CT of the right ear demonstrates a large congenital cholesteatoma filling the middle ear cavity with subtle long process of…



Axial bone CT of the right ear reveals a medial epitympanic congenital cholesteatoma eroding the medial head of the malleus and short proces…



Coronal bone CT in the same patient shows CMEC eroding the long process of the incus and filling the oval window niche. The tympanic segment of CNVI…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1026.

TEMPORAL BONE & SKULL BASE · TRAUMA

Skull Base Fractures

IMAGING FINDINGS

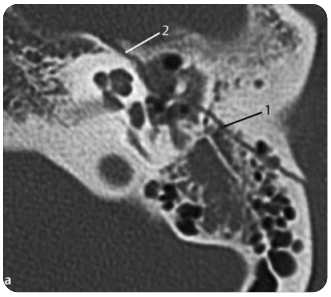
- Fracture line through the temporal bone or skull base; air in the middle ear or cranium
- Otic capsule sparing (most) or violating
- Fluid in the middle ear and mastoid

TOP DIFFERENTIALS

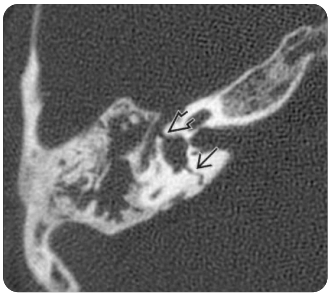
- Sutures, fissures and small canals (pseudofractures)
- Vascular grooves

CHECKLIST

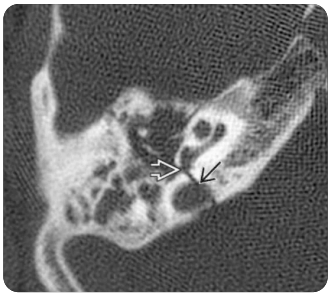
- Ossicles, facial canal (geniculate), tegmen and CSF leak
- Carotid canal involved: CT angiography



CT, axial. Patient with head trauma showing a fracture line through the mastoid (1) extending just anterior to the cochlea (2). Luxation of the…



Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…



Axial bone CT more inferiorly in the same patient shows typical inferior extension of the fracture across the jugular foramen, into the round windo…



Axial bone CT shows a chronic transverse fracture line extending through the otic capsule to the basal turn of the cochlea, where abnormal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

INNER EAR & IAC · CONGENITAL

Malformations of the Inner Ear

IMAGING FINDINGS

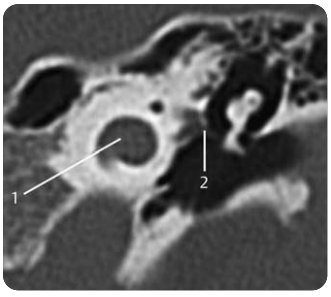
- Abnormal cochlea, vestibule or canals on CT and high-resolution T2
- IP-II: cystic apex, normal basal turn, large vestibular aqueduct
- Common cavity, cochlear hypoplasia, IP-I and IP-III by appearance

TOP DIFFERENTIALS

- Labyrinthitis ossificans
- Normal variant of canal size

CHECKLIST

- Cochlear nerve present on oblique sagittal T2
- Cochlear nerve canal width; gusher risk (IP-I, IP-III)



CT, coronal. This coronal view of the cochlea does not show any bony modiolus (1). This empty cochlea is suggestive of an incomplete partition…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1102.

INNER EAR & IAC · CONGENITAL

Vestibular Aqueduct, Enlargements

IMAGING FINDINGS

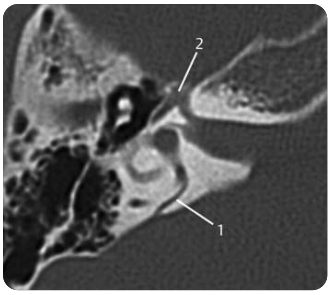
- Vestibular aqueduct wider than about 1 mm at the midpoint or 2 mm at the operculum
- Large endolymphatic sac on T2
- Often with IP-II cochlear deficiency

TOP DIFFERENTIALS

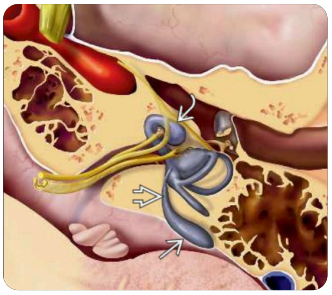
- Normal aqueduct (compare with the posterior canal)
- IP-I cystic cochleovestibular malformation

CHECKLIST

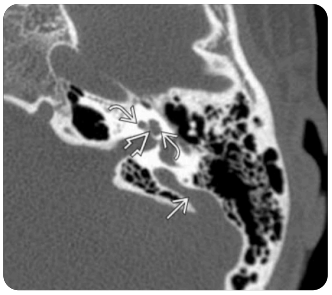
- Look at both ears: often bilateral
- Pendred syndrome association



CT, axial. The normal vestibular aqueduct (1) in its passage from the vestibule to its junction with sinuses in the region of the sigmoid sinus. Ther…



In left inner ear, large endolymphatic sac epidural and intraosseous components are shown. Cochlea is malformed, with absent septation…



Axial temporal bone CT in a 15-year-old boy with sensorineural hearing loss

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104



Axial temporal bone CT in a 17-year-old girl with Pendred syndrome shows an LVA and absent septation between the middle and apical cochlear turns with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1104.

INNER EAR & IAC · BONE DISEASE

Labyrinthitis Ossificans (Ossification after Meningitis)

IMAGING FINDINGS

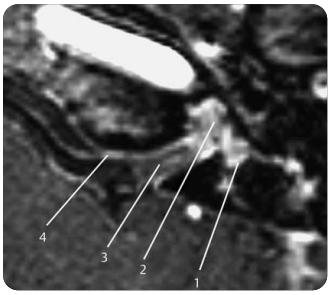
- Loss of the bright fluid signal of the cochlea on high-resolution T2
- Later: bone in the basal turn and canals on CT
- Early: enhancement of the labyrinth

TOP DIFFERENTIALS

- Labyrinthine or cochlear aplasia
- Cochlear otosclerosis
- Intralabyrinthine schwannoma

CHECKLIST

- After meningitis: urgent MRI and cochlear implantation
- Report patency of the basal turn



MRI, T1-weighted with gadolinium enhancement, axial. There is enhancement of the vestibule (1), cochlear contents (2), and the…

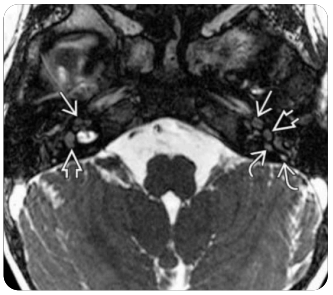


Axial bone CT in a 2-year-old boy with bilateral sensorineural hearing loss

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116



Axial bone CT in the same patient demonstrates subtotal ossification of the anterior and midportion of the lateral semicircular canal. The…



Axial thin section T2WI MR in a 2-year-old with bacterial meningitis 3 weeks prior shows diffuse, bilateral decrease in normal hyperintense inner ear fluid…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116.

TEMPORAL BONE & SKULL BASE · TUMOURS

Pseudotumors in the Apex of the Petrous Bone

IMAGING FINDINGS

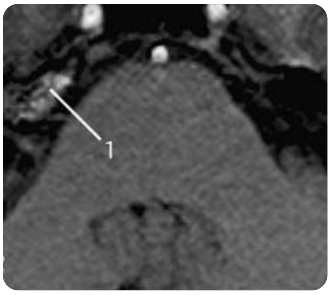
- Asymmetric marrow: bright on T1, suppresses with fat saturation, not expanded
- Trapped fluid: bright on T2, intact septa
- Cholesterol granuloma: bright on T1 and T2, expansile

TOP DIFFERENTIALS

- Cholesteatoma or epidermoid (DWI bright)
- Petrous apicitis (destruction, enhancement)
- Carotid aneurysm

CHECKLIST

- Compare with the other apex
- CT for expansion and cortex



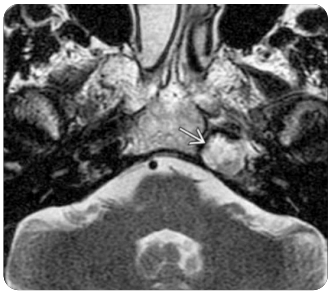
MRI, T1-weighted, axial. T1-weighted image shows that the lesion (1) has high signal intensity compared with brain tissue (2). The contralateral...



Axial bone CT reveals an expansile cholesterol granuloma in the left PA with marginal bone dehiscence present. The right PA is well...



Axial T1WI unenhanced MR in the same patient shows characteristic high signal of PA-CG. If no fat saturation is applied to the enhanced sequences...



Axial T2WI MR in the same patient demonstrates the lesion has inhomogeneous high signal. Cholesterol granuloma contains "old...

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152.

INNER EAR & IAC · TUMOURS

Vestibular Schwannoma

IMAGING FINDINGS

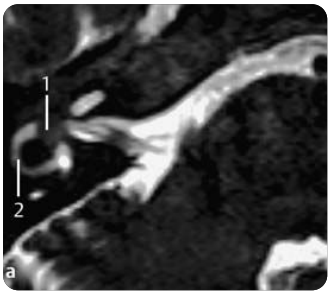
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

TOP DIFFERENTIALS

- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



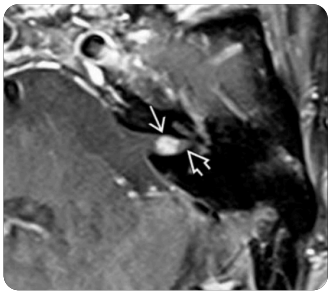
MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul...



Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by...



Axial T1WI C+ MR demonstrates a large CPAIAC vestibular schwannoma compressing the middle cerebellar peduncle and cerebellar hemisphere....



Axial T1WI C+ FS MR in a patient with left sensorineural hearing loss shows a small enhancing vestibular schwannoma within the IAC with a 3-...

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.

INNER EAR & IAC · INFECTION & INFLAMMATION

(Epi)dermoid Cyst

IMAGING FINDINGS

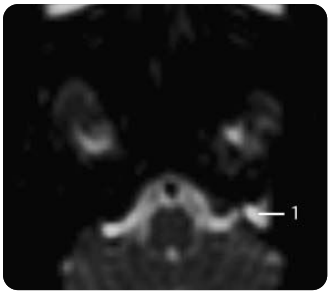
- CSF-like lesion that insinuates around nerves and vessels
- Incomplete suppression on FLAIR
- Bright on diffusion; no enhancement

TOP DIFFERENTIALS

- Arachnoid cyst (follows CSF on every sequence, DWI dark)
- Cystic schwannoma

CHECKLIST

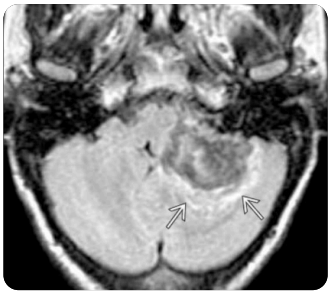
- Encasement of cranial nerves and vessels
- Extent across cisterns before surgery



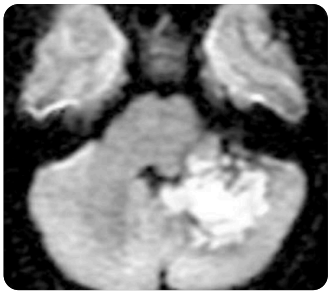
Patient with progressive and asymmetrical perceptive hearing loss.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.9 a



Axial CECT shows a large CPA epidermoid cyst. Note that this non-enhancing low-density lesion appears to invade the left cerebellar…



Axial FLAIR MR of the same patient shows "incomplete" or partial nulling of the signal of this large epidermoid cyst. Associated high signal along th…



Axial DWI MR in the same patient reveals the expected high signal from epidermoid cyst diffusion restriction. DWI sequence allows straightforward…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1202.

INNER EAR & IAC · TUMOURS

Meningioma of the Cerebellopontine Angle

IMAGING FINDINGS

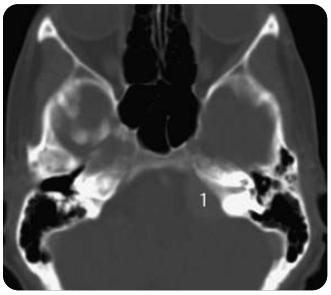
- Broad-based dural mass, obtuse angle with the petrous bone
- Eccentric to the porus; dural tail
- Calcification and hyperostosis; uniform enhancement

TOP DIFFERENTIALS

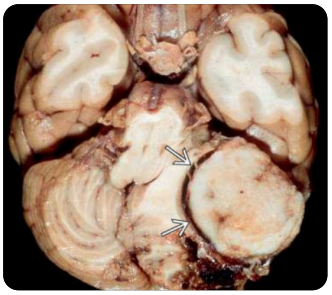
- Vestibular schwannoma
- Epidermoid cyst
- Dural metastasis

CHECKLIST

- Extension into the IAC, jugular foramen and Meckel cave
- Venous sinus involvement



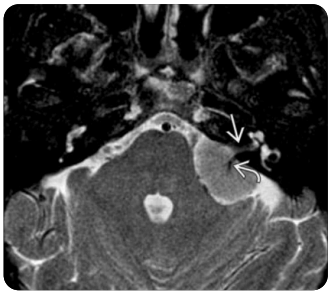
CT, axial. Just before the birth of her child, screening with CT without contrast enhancement suggested a lesion in the left CPA (1). The size of…



Gross pathologic section viewed from below shows a large CPA meningioma with a broad dural base compressing the cerebellum. The specimen…



Axial T1WI C+ FS MR through the IAC shows a meningioma overlying the porus acusticus. Note the dural tail extending along the temporal bone…



Axial T2WI FS MR in the same patient reveals a high-velocity flow void representing a dural artery feeder penetrating the meningioma core. Lo…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1228.

TEMPORAL BONE & SKULL BASE · TUMOURS

Glomus Jugulare Tumor (Paraganglioma)

IMAGING FINDINGS

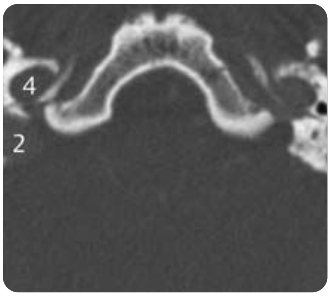
- Jugular foramen mass with permeative, moth-eaten bone erosion
- "Salt and pepper" on T1 with intense enhancement
- Spread superolaterally into the middle ear

TOP DIFFERENTIALS

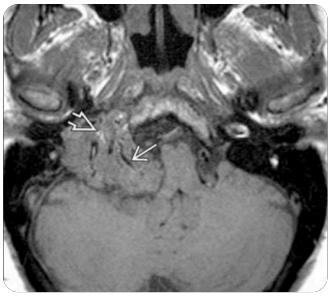
- Jugular foramen schwannoma (smooth scalloping)
- Meningioma (sclerosis, dural tail)
- Dehiscent jugular bulb or metastasis

CHECKLIST

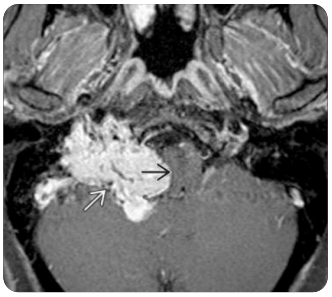
- Carotid, jugular vein and lower cranial nerves
- Multicentric paragangliomas: scan the neck



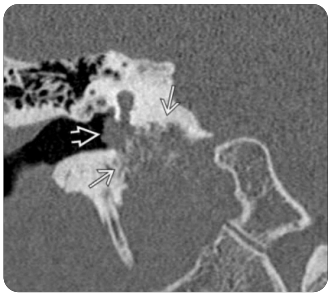
Patient with pulsating tinnitus, without any pathology behind the eardrum.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.20 a



Axial T1WI MR shows a large mass arising from the jugular foramen with multiple areas of salt & pepper. The "salt" represents blood products or…



Axial T1WI C+ FS MR in the same patient shows the classic intense enhancement of this large vascular tumor. There is intracranial extension…



Coronal T-bone CT of the right ear shows the classic permeative-destructive margins of GJP. Note the typical vector of spread superolateral…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892.

NOSE & NASAL CAVITY · CONGENITAL

Meningoencephalocele

IMAGING FINDINGS

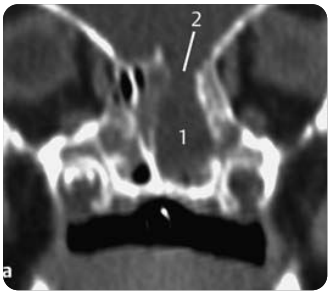
- Bony defect in the cribriform plate, lateral lamella, ethmoid roof or sphenoid
- Soft tissue or fluid through the defect
- MRI: CSF, meninges or brain in continuity with the intracranial contents

TOP DIFFERENTIALS

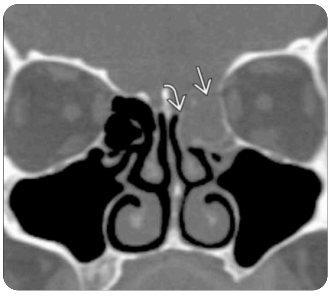
- Skull base defect without leak
- Sinus opacification from mucus

CHECKLIST

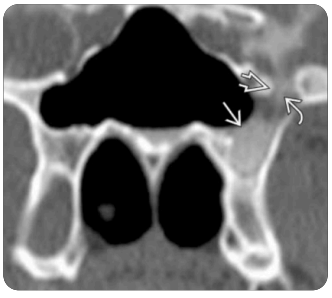
- Lateral sphenoid recess and multiple defects
- Signs of raised intracranial pressure: empty sella



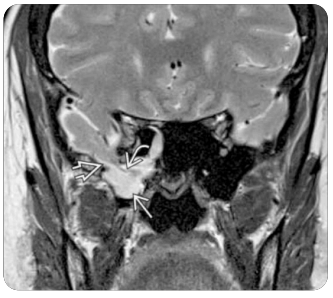
CT, coronal. The left nasal cavity is filled with a soft-tissue mass (1) that seems to be connected to brain through a defect in the roof of the…



Coronal bone CT shows large bony defect in left ethmoid roof, lateral to insertion of middle turbinate. Because there is complete opacification of the…



Coronal bone CT after intrathecal contrast shows CSF in left sphenoid chamber, bone defect, and contrast extending through defect. CT…



Coronal T2WI MR reveals lateral sphenoid sinus filled with CSF. Note the osseous defect in lateral sphenoid roof, and brain herniating through…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Dural Sinus Thrombosis

IMAGING FINDINGS

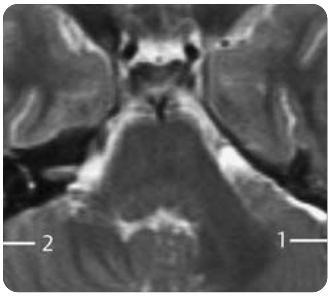
- Filling defect in the sigmoid or transverse sinus on contrast CT or MR venography
- Empty delta sign; dense sinus on non-contrast CT
- Adjacent mastoiditis with bone erosion over the sinus

TOP DIFFERENTIALS

- Arachnoid granulation (round, CSF signal)
- Hypoplastic transverse sinus
- Slow flow artefact

CHECKLIST

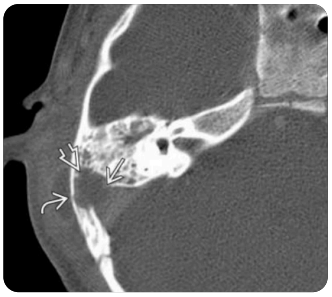
- Extension to the jugular vein and other sinuses
- Venous infarction and raised intracranial pressure



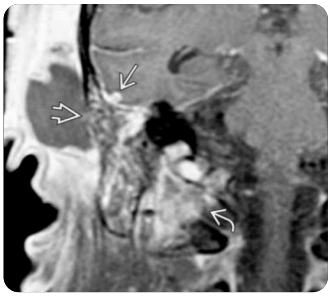
MRI, T2-weighted, axial. Otomastoiditis can spread intracranially, resulting in dural sinus thrombophlebitis, which may compromise the sigmoid and…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…



Axial bone CT of the same patient shows loss of mastoid trabecula and a dehiscent sigmoid plate diagnostic of coalescent otomastoiditis. Subtle…



Coronal T1WI C+ FS MR reveals mastoid enhancement with a large periauricular abscess. Lateral mastoid is focally dehiscent with thick and enhancing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Maxillary Sinusitis

IMAGING FINDINGS

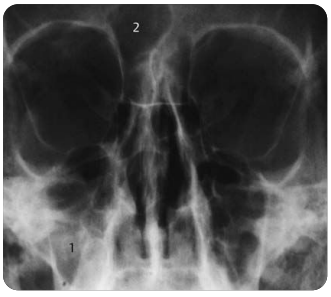
- Air–fluid level, frothy or bubbly secretions
- Mucosal thickening
- Chronic: sclerotic thickened walls (osteitis)

TOP DIFFERENTIALS

- Retention cyst or polyp
- Post-traumatic blood
- Odontogenic source (periapical lucency, oroantral fistula)

CHECKLIST

- Look at the teeth and the ostiomeatal unit
- Contrast CT only for complications



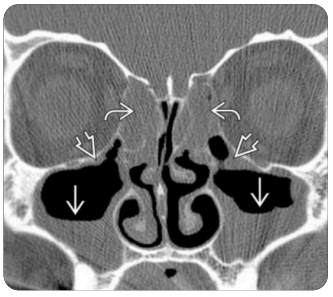
Caldwell view. This 26-year-old patient had pain in the right maxillary region. The right maxillary sinus seems to be opacified in the lower part (1). The lef…



Axial NECT in patient with purulent nasal discharge and headache shows patchy bilateral ethmoid sinus disease, an air-fluid level in the right sphenoid…



Coronal reconstruction NECT demonstrates bubbly, frothy secretions in sphenoid sinuses in a patient with clinical diagnosis of ABR…



Coronal bone CT shows acute bilateral maxillary and ethmoid sinusitis. Note airfluid levels and mucosal thickening in maxillary sinuses and complete…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 688.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Adenoid Hypertrophy

IMAGING FINDINGS

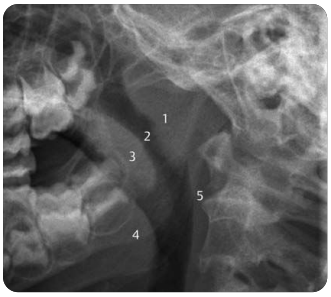
- Symmetric soft tissue in the nasopharyngeal roof and posterior wall
- Striated pattern on T2 with uniform enhancement
- Narrowed nasopharyngeal airway on a lateral film

TOP DIFFERENTIALS

- Nasopharyngeal carcinoma or lymphoma in an adult
- Thornwaldt cyst
- Juvenile angiofibroma

CHECKLIST

- In adults, asymmetric or bulky adenoids need endoscopy
- Middle ear effusions



Plain radiograph for adenoid evaluation.

Dunnebier, Imaging for Otolaryngologists, Fig. 7.2

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Nasal Polyps

IMAGING FINDINGS

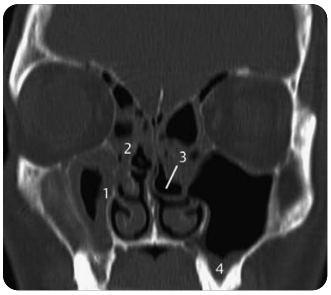
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

TOP DIFFERENTIALS

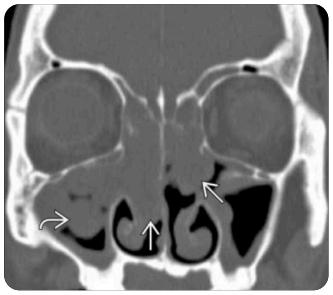
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

CHECKLIST

- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist

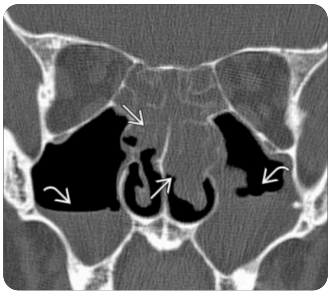


CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…

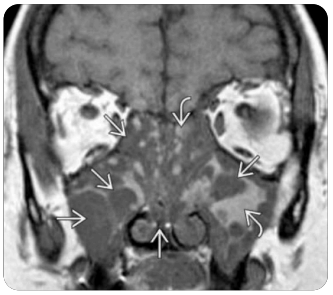


Coronal bone CT shows the classic appearance of sinonasal polyposis

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708



Coronal bone CT in a patient with polyposis and new onset of acute sinusitis symptoms shows polyps in the nasal cavity occluding the middle…



Coronal T1WI MR shows multiple intermediate signal intensity polyps filling the nasal cavity maxillary, and ethmoid sinuses. Trapped secretions…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Noninvasive Fungal Sinusitis

IMAGING FINDINGS

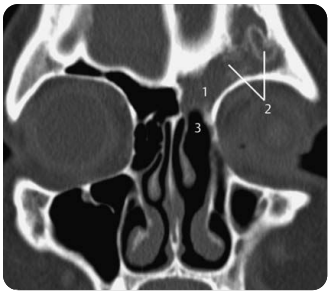
- Fungal ball: single sinus with dense contents and calcification, sclerotic walls
- Allergic fungal: expanded sinuses with central hyperdensity and low-density rim
- Very dark on T2

TOP DIFFERENTIALS

- Polyposis
- Mucocele
- Inspissated secretions

CHECKLIST

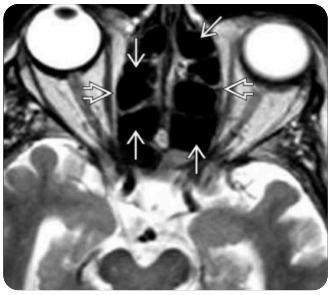
- Lamina papyracea and skull base remodelling
- Usually unilateral or asymmetric



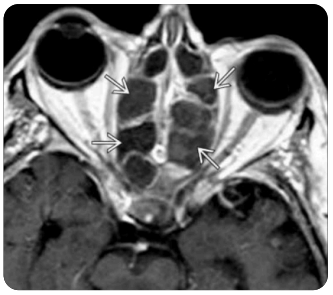
Patient with constant left frontal sinus pressure. CT, coronal (a), sagittal (b), and axial ©. There is complete opacification of the left frontal sinus…



Axial NECT shows unilateral involvement of allergic fungal sinusitis. The involved sinuses are expanded, with centrally dense inspissated…



Axial T2WI MR in a patient with bilateral AFS shows diffuse hypointense signal within the involved ethmoid air cells. There is mild sinus expansion with…



Axial T1WI C+ MR in the same patient confirms that the ethmoid sinuses are not aerated, but are in fact completely opacified. The surrounding mucosa…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 700.

NOSE & NASAL CAVITY · CONGENITAL

Choanal Atresia

IMAGING FINDINGS

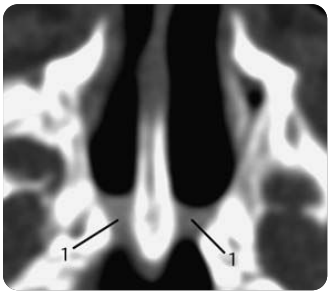
- Narrowed posterior choanae
- Thickened vomer; medially bowed posterior maxilla
- Bony, membranous or mixed closure; fluid in the nose

TOP DIFFERENTIALS

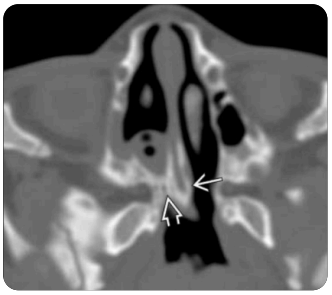
- Choanal stenosis
- Pyriform aperture stenosis
- Nasolacrimal duct mucocele

CHECKLIST

- CT after suction and decongestion
- CHARGE: look at the inner ears



CT, axial. Both choanal areas demonstrate narrowing of the bony passage, but congenital fibrous closure (1) is the cause of the distres…



Axial bone CT through the upper choana in a child with mixed bony/osseous choanal atresia shows osseous choanal obstruction…



Axial bone CT through the mid choana in the same patient shows membranous atresia bridging the narrowed inferior aspect of the…



Axial bone CT in a child with CHARGE syndrome demonstrates bilateral choanal obstruction secondary to linear membranes and mildly thickene…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Orbital Cellulitis

IMAGING FINDINGS

- Stranding of the postseptal fat
- Swollen extraocular muscles; proptosis
- No discrete collection

TOP DIFFERENTIALS

- Subperiosteal abscess
- Invasive fungal infection
- Idiopathic orbital inflammation

CHECKLIST

- Separate preseptal from postseptal
- Repeat imaging if not improving



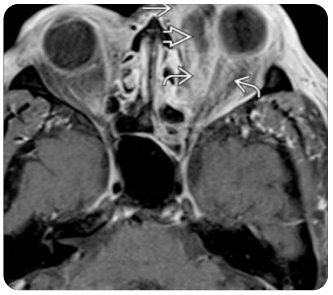
CT, axial with intravenous contrast. Bilateral opacification of the ethmoid (1). The lamina papyracea is intact without signs of periosteal…



Axial CECT in a patient with facial impetigo shows edema and enhancement of the preseptal soft tissues with normal appearance of th…



Axial STIR MR in a teenage girl shows infiltrating signal representing both preseptal and postseptal cellulitis secondary to ethmoid sinusitis.…



Axial T1-weighted post-contrast MR in a patient with recent penetrating facial trauma shows extensive preseptal and intraorbital infiltration and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 814.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Subperiosteal Orbital Abscess

IMAGING FINDINGS

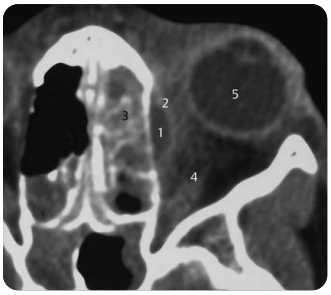
- Lentiform rim-enhancing collection along the lamina papyracea
- Medial rectus and globe displaced laterally
- Adjacent ethmoid opacification; diffusion restriction

TOP DIFFERENTIALS

- Orbital cellulitis (phlegmon)
- Idiopathic orbital inflammation
- Mucocele

CHECKLIST

- Size, posterior extent and vision
- Superior ophthalmic vein and cavernous sinus



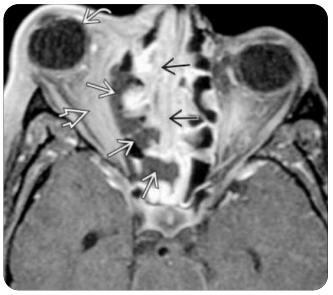
CT, axial with intravenous contrast enhancement. The CT scan shows a subperiosteal orbital abscess (1) with typical rim enhancement (2). The…



Axial CECT shows asymmetric opacification of the left ethmoid sinuses with a large subperiosteal abscess extending into the medial…



Coronal T1WI C+ FS MR in a patient status post endoscopic sinus surgery shows phlegmonous enhancement in the medial right extraconal fat with a…



Axial T1WI C+ FS MR shows a large subperiosteal abscess extending through the lamina papyracea with findings of acute sinusitis and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 810.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Invasive Fungal Sinusitis

IMAGING FINDINGS

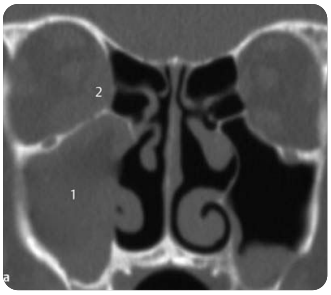
- Unilateral mucosal thickening with fat stranding outside the sinus
- Subtle bone erosion; loss of fat in the pterygopalatine fossa or orbit
- Non-enhancing mucosa ("black turbinate") on MRI

TOP DIFFERENTIALS

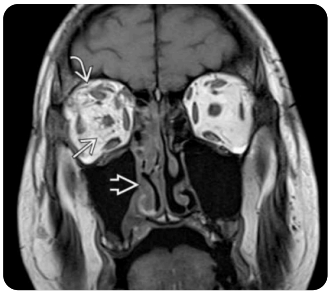
- Acute bacterial sinusitis with complication
- Lymphoma or carcinoma

CHECKLIST

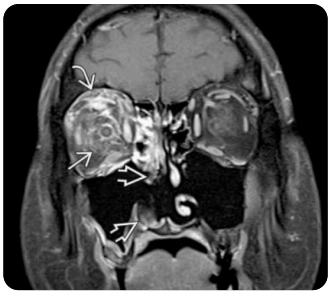
- Immunocompromised patient: endoscopy and biopsy without delay
- Orbit, cavernous sinus and brain on MRI



CT, coronal, bone-tissue setting. The bone-tissue setting reveals complete opacification of the right maxillary sinus (1), as well as very doubtful…



Coronal T1WI MR in patient with chronic myelogenous leukemia complicated by Rhizopus acute invasive fungal rhinosinusitis…



Coronal T1 C+ FS shows mucosal nonenhancement of right inferior & middle turbinates

Koch et al., Diagnostic Imaging: Head…



Axial T1WI MR in a patient with chronic myelogenous leukemia complicated by Rhizopus AIFRS shows ill-defined softtissue infiltration of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 704.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Mucocele

IMAGING FINDINGS

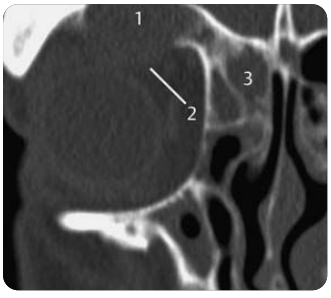
- Expanded, airless sinus with smooth remodelled walls
- Variable density and signal with protein content
- Rim enhancement only

TOP DIFFERENTIALS

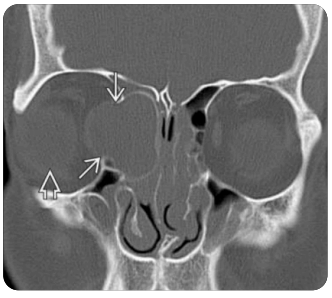
- Allergic fungal sinusitis
- Polyposis
- Tumour (solid enhancement, destruction)

CHECKLIST

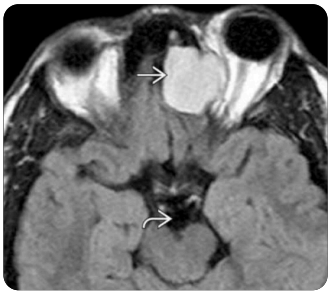
- Orbit and skull base erosion
- Frontal recess obstruction as the cause



CT, coronal. The left frontal sinus is opacified with expansion of its inferior bony outline through the roof of the orbit (1). Although this is a bone-…



Coronal bone CT shows the typical features of a right ethmoid mucocele. The sinus is opacified, and there is remodeling of the surrounding bony…



Axial FLAIR MR in a patient with proptosis and diplopia shows a homogeneous, expansile left ethmoid mucocele. Note that the material…



Sagittal T1 C+ MR demonstrates increased T1 signal in the mucocele, likely related to elevated protein content. There is no central…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716.

ORBIT & SINUS RELATIONS · TRAUMA

Blow-out Fracture

IMAGING FINDINGS

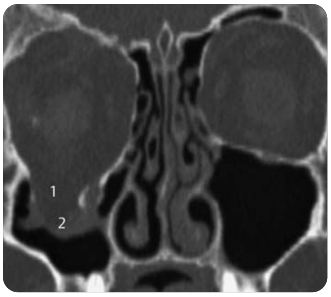
- Fracture of the orbital floor or medial wall with the rim intact
- Herniated fat and muscle; rounded inferior rectus
- Blood in the maxillary sinus; orbital emphysema

TOP DIFFERENTIALS

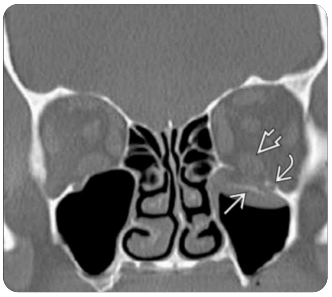
- Dehiscent lamina papyracea
- Old fracture

CHECKLIST

- Muscle entrapment, especially the trapdoor fracture in children
- Defect size (enophthalmos) and retrobulbar haemorrhage



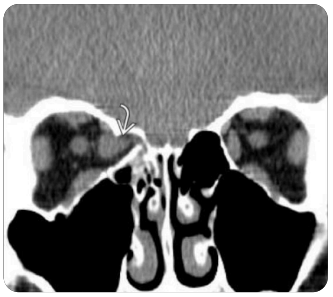
CT, coronal. Traumatic disruption of the bony floor of the orbit (1), and prolapse of orbital fat into the maxillary sinus (2). The eyeball was at a lower…



Coronal bone CT demonstrates a mildly depressed left orbital floor orbital blowout fracture
Koch et al., Diagnostic Imaging: Head…



Coronal NECT reveals a chronic right orbital floor OBF with herniation of a small volume of fat and a portion of the inferior rectus muscle through the…



Coronal soft tissue NECT shows deformity of the right medial rectus muscle, entering the osseous defect of a lamina papyracea fracture.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 980.

NOSE & NASAL CAVITY · TUMOURS

Inverted Papilloma

IMAGING FINDINGS

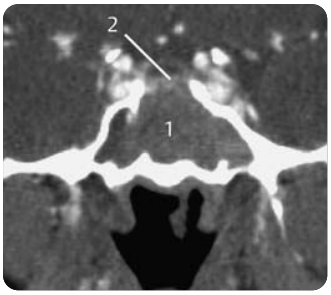
- Unilateral mass on the lateral nasal wall near the middle meatus
- Focal hyperostosis at the attachment
- Convoluted cerebriform pattern on T2 and post-contrast MRI

TOP DIFFERENTIALS

- Antrochoanal or solitary polyp
- Squamous cell carcinoma (in up to about 10%)
- Juvenile angiofibroma

CHECKLIST

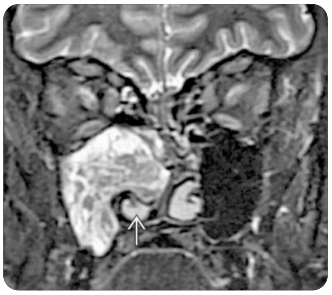
- MRI separates tumour from secretions
- Bone destruction raises concern for carcinoma



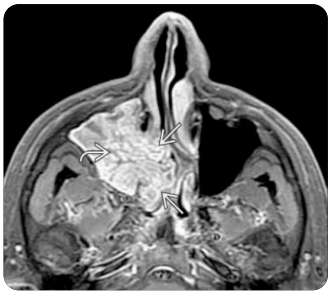
Patient operated before because of nasal polyps. Histologic evaluation revealed inverted papilloma. Follow-up scans are shown.…



Axial bone CT shows a mass in the nasal cavity along the lateral wall near the middle meatus. The maxillary sinus is opacified, but it is difficult to…



Coronal T2 FS MR shows characteristic features on an inverted papilloma that involves the right maxillary sinus and nasal cavity. The lesion has a…



Axial T1 C+ FS MR shows classic features of an inverted papilloma. The lesion is centered at the middle meatus with nasal cavity and antral…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 744.

NOSE & NASAL CAVITY · TUMOURS

Angiofibroma, Juvenile

IMAGING FINDINGS

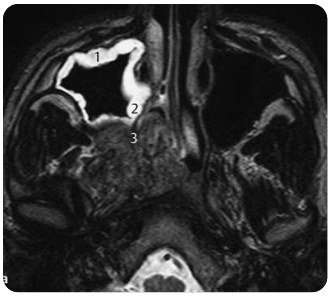
- Enhancing mass at the sphenopalatine foramen in an adolescent male
- Widened pterygopalatine fossa; bowing of the posterior maxillary wall
- Flow voids; extension to the sphenoid, infratemporal fossa and skull base

TOP DIFFERENTIALS

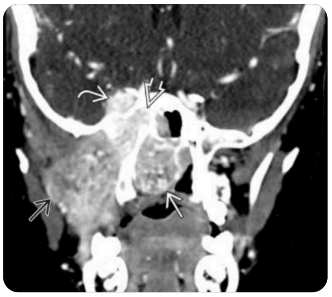
- Hypervascular polyp
- Rhabdomyosarcoma
- Antrochoanal polyp

CHECKLIST

- No biopsy: angiography and embolisation before surgery
- Vidian canal and cavernous sinus spread



A young man with obstruction and bloodtinged nasal discharge from the right side of the nose.
Dunnebier, Imaging for…



Coronal CECT in a 13-year-old shows typical growth pattern of a large, hypervascular angiofibroma occluding the nasopharynx, extending into righ…



Axial T1 C+ FS MR in a 15-year-old with nasal obstruction shows enhancing angiofibroma filling the left nasal cavity and extending into the…



Axial T1 C+ FS MR in the same patient shows a few intralesional high-flow vessels and extension of angiofibroma into the widened PPF, destruction of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Nasopharyngeal Carcinoma

IMAGING FINDINGS

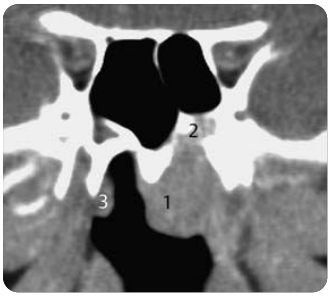
- Asymmetric mass in the fossa of Rosenmüller
- Parapharyngeal and skull base spread; clival marrow replaced on T1
- Retropharyngeal and level II and V nodes

TOP DIFFERENTIALS

- Adenoidal lymphoid hyperplasia (symmetric, striated)
- Lymphoma
- Minor salivary gland tumour

CHECKLIST

- MRI for staging: skull base, cavernous sinus, perineural spread
- Middle ear effusion from Eustachian tube obstruction



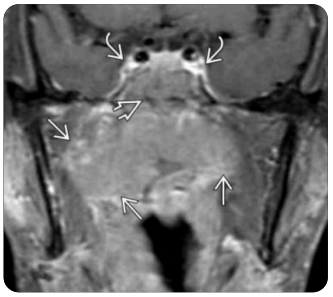
CT, coronal with intravenous contrast enhancement. Note the soft-tissue mass in the roof of the nasopharynx of the left side (1), with invasion in the…



Coronal NECT of paranasal sinuses, performed in a 62-year-old Asian man for evaluation of nasal stuffiness and epistaxis, demonstrates an…



Sagittal T1WI MR in the same patient reveals a large soft tissue mass in the nasopharynx extending superiorly to the sphenoid and sella. There is…



Coronal T1WI C+ FS MR reveals moderate enhancement of a bulky mass filling the nasopharynx with an indistinct right lateral margin.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Tonsillar Abscess

IMAGING FINDINGS

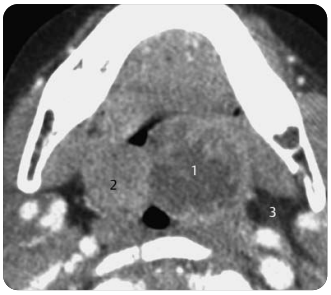
- Rim-enhancing fluid collection in or around the tonsil
- Swollen tonsil and pharyngeal wall
- Spread to parapharyngeal, masticator or carotid space

TOP DIFFERENTIALS

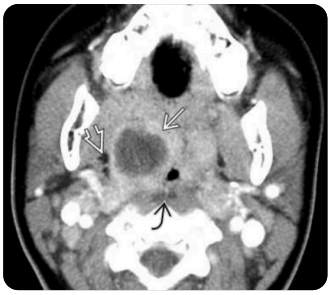
- Tonsillitis without abscess (phlegmon)
- Tonsillar retention cyst
- Retropharyngeal abscess; tonsillar carcinoma

CHECKLIST

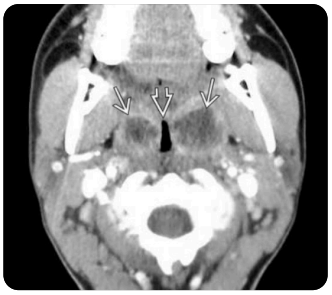
- Airway
- Internal jugular vein thrombosis and carotid



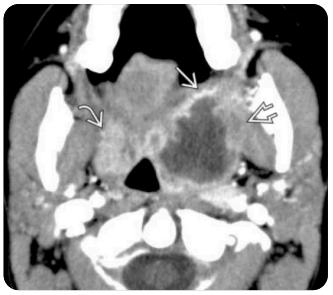
CT with intravenous contrast enhancement, axial. The left tonsil (1) is much larger than the contralateral side (2) with central hypodensities…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…



Axial CECT reveals bilateral tonsillar abscesses, the left larger than the right. Note the mass effect on both PPS but no extension of inflammatory…



Axial CECT demonstrates left tonsillar and peritonsillar abscess with extension through the capsule into the posterior buccal space and medial…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Tonsillar Carcinoma

IMAGING FINDINGS

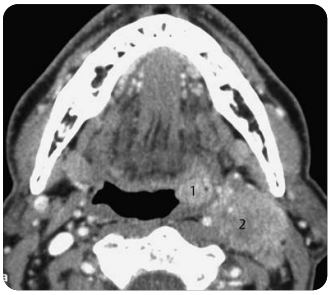
- Asymmetric enhancing tonsillar mass, sometimes small
- Spread to soft palate, tongue base and parapharyngeal space
- Cystic or necrotic level II nodes (HPV)

TOP DIFFERENTIALS

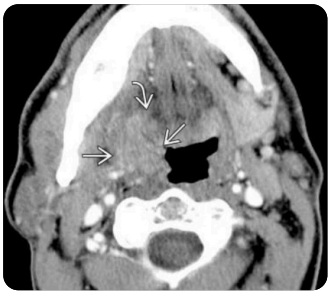
- Tonsillar lymphoid hyperplasia
- Lymphoma
- Peritonsillar abscess

CHECKLIST

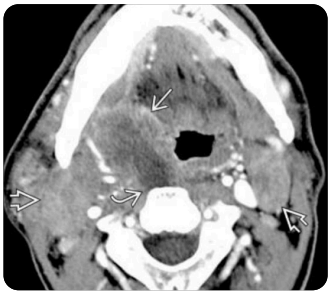
- Medial pterygoid, mandible and carotid
- Both sides of the neck



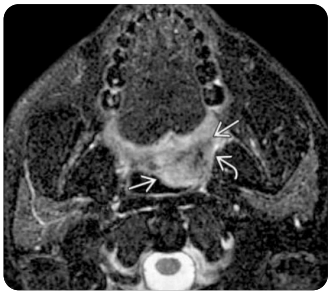
CT with intravenous contrast enhancement, axial. In the left tonsillar fossa, a small and enhancing tumor is visible (1), with enlarged level II lymph…



Axial CECT in alcoholic smoker presenting with multiple right neck masses shows an enlarged right palatine tonsil mass…



Axial CECT, same patient 1 year later, shows dramatic progression of disease. Stage is now T4a N2c M1. The tumor is much larger, extending into…



Axial T2WI FS MR in a 50-year-old presenting with a palatine tonsil mass shows heterogeneous exophytic left tonsillar tumor. Well-defined lateral…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 492.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Oropharyngeal Tumor, Base of Tongue

IMAGING FINDINGS

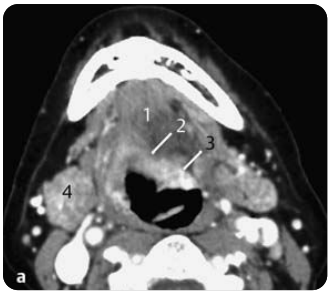
- Asymmetric fullness of the lingual tonsil with enhancement
- Spread to vallecula, pre-epiglottic space and floor of mouth
- Level II nodes, often bilateral

TOP DIFFERENTIALS

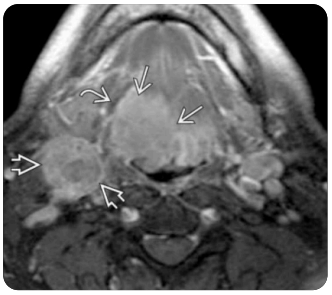
- Lingual tonsil lymphoid hyperplasia
- Lymphoma
- Thyroglossal or vallecular cyst

CHECKLIST

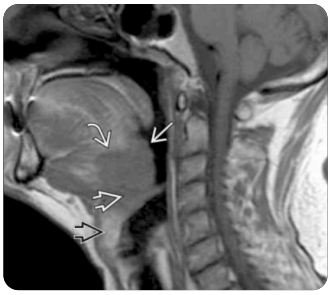
- Crossing the midline and neurovascular bundles
- Pre-epiglottic and laryngeal invasion



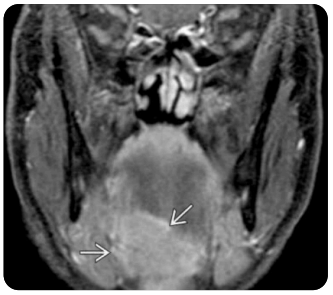
CT with intravenous contrast enhancement, axial. Axial slice at the level of the mandible and floor of the mouth (1). At clinical evaluation, an…



Axial T1WI C+ FS MR in a 58-year-old alcoholic presenting with persistent right neck mass after dental procedure demonstrates enlarged,…



Sagittal T1WI MR in the same patient shows fullness at the right base of tongue with a tumor replacing the lingual tonsillar tissue and infiltrating…



Coronal T1WI C+ FS MR shows an asymmetric mass at the right base of tongue in the same patient. This was found to be an invasive SCCa and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 488.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Abscess of the Neck

IMAGING FINDINGS

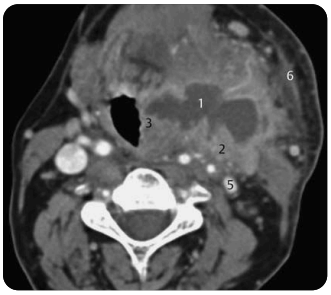
- Low-density collection with a well-formed enhancing rim
- Surrounding cellulitis and fat stranding
- The source: tooth, tonsil, node, stone or foreign body

TOP DIFFERENTIALS

- Phlegmon (no fluid centre)
- Necrotic metastatic node
- Infected congenital cyst

CHECKLIST

- Airway, mediastinum and jugular vein
- Which space: decides the drainage route



CT with intravenous contrast enhancement, axial.A large abscess (1) with a thick, contrast-enhanced capsule (2). The abscess is extendin…

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Parapharyngeal Abscess

IMAGING FINDINGS

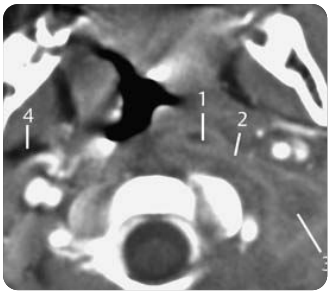
- Collection in the parapharyngeal fat, usually from the tonsil
- Medial bulge of the pharyngeal wall
- Trismus from medial pterygoid involvement

TOP DIFFERENTIALS

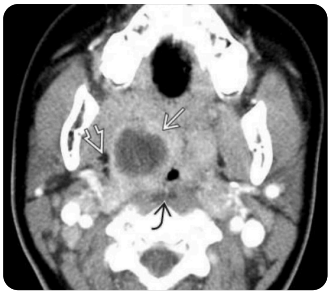
- Peritonsillar abscess
- Retropharyngeal abscess
- Deep lobe parotid infection

CHECKLIST

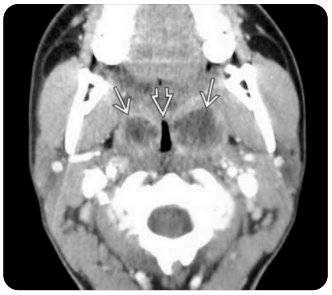
- Carotid space and jugular vein (Lemierre)
- Spread to the mediastinum



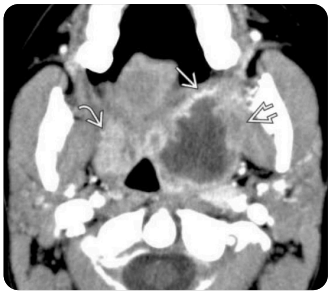
CT with intravenous contrast enhancement, axial. Note the bulging left pharyngeal wall, as a result of a parapharyngeal abscess (1) which is…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…



Axial CECT reveals bilateral tonsillar abscesses, the left larger than the right. Note the mass effect on both PPS but no extension of inflammatory…



Axial CECT demonstrates left tonsillar and peritonsillar abscess with extension through the capsule into the posterior buccal space and medial…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Cervical Lymphadenitis

IMAGING FINDINGS

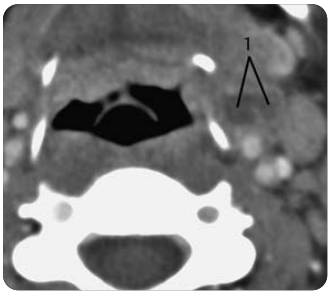
- Enlarged nodes with preserved fatty hilum (reactive)
- Suppurative node: thick rim around a fluid centre with surrounding inflammation
- Matted necrotic nodes in tuberculosis

TOP DIFFERENTIALS

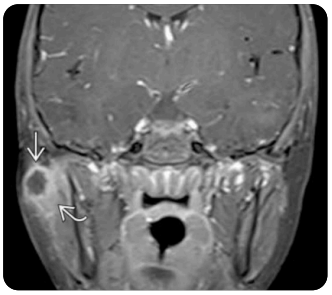
- Metastatic squamous carcinoma
- Second branchial cleft cyst
- Atypical mycobacteria

CHECKLIST

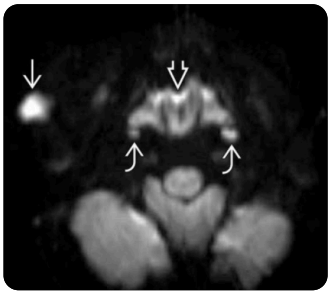
- Ultrasound first in children
- Progression to abscess



CT with intravenous contrast enhancement, axial. Various enlarged level III lymph nodes are present near the internal carotid artery and jugular…



Coronal T1 C+ FS MR in an 8-year-old boy with right cheek swelling shows a peripherally enhancing, centrally necrotic intraparotid mass with…



Axial DWI trace in the same patient shows the parotid mass to have central reduced diffusivity, typical of suppurative adenitis. Note reactive…



Axial CECT demonstrates a well-defined, low-density submandibular mass with thin peripheral rim enhancement. Suppurative node is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 314.

NECK: TRIANGLES, FASCIA & NODES · TUMOURS

Lymphoma, Non-Hodgkin

IMAGING FINDINGS

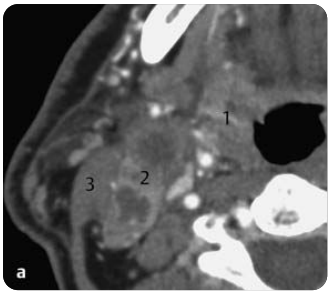
- Large, homogeneous, non-necrotic nodes in many chains
- Waldeyer ring involvement
- Strong diffusion restriction

TOP DIFFERENTIALS

- Reactive nodes
- Tuberculosis
- Metastatic carcinoma

CHECKLIST

- Extranodal sites: tonsils, sinuses, orbit, thyroid
- Core or excision biopsy, not FNA alone



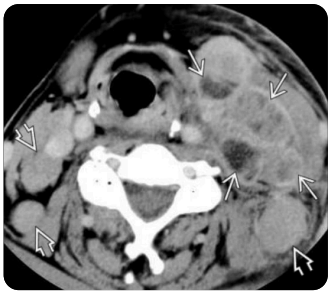
CT with intravenous contrast enhancement, axial. On this slice, an area of different structure with enhancement is seen at the base of…



Axial CECT shows multiple large, round, solid lymph nodes in every chain of the suprahyoid neck. Bilateral nodes are evident in levels IIA, IIB, and…



Axial CECT in a different patient shows nodal and tonsillar lymphoma. Level II nodal mass is partially necrotic, with extranodal infiltration of fat and…



Axial CECT demonstrates massive left nodal conglomerate, some with necrosis. Surrounding induration and stranding of fat in left neck suggests…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 336.

NECK VESSELS & NERVES · TUMOURS

Carotid Body Paraganglioma

IMAGING FINDINGS

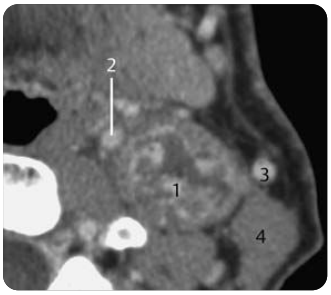
- Avidly enhancing mass in the carotid bifurcation
- Splaying of the internal and external carotid arteries
- Flow voids ("salt and pepper") on MRI

TOP DIFFERENTIALS

- Vagal paraganglioma (pushes ICA forwards)
- Carotid space schwannoma
- Jugulodigastric node; pseudoaneurysm

CHECKLIST

- Shamblin class: degree of carotid encasement
- Look for multiple paragangliomas



Patient with a slowly progressive lesion in the neck.

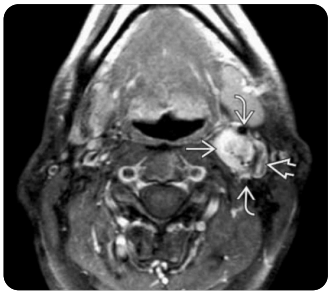
Dunnebier, Imaging for Otolaryngologists, Fig. 9.10 a



Axial CECT shows a classic carotid body paraganglioma with avid, fairly uniform enhancement. Notice the clear definition of the tumor sitting in the…



Axial T1WI MR shows a rounded mass in the left carotid space, between 2 flow voids representing the ICA and ECA. Small internal foci of signal void…



Axial T1WI C+ FS MR in the same patient shows intense enhancement of a solid mass between the splayed branches of the carotid artery. The…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 140.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Branchial Cleft Cyst

IMAGING FINDINGS

- Cyst at the angle of the mandible in front of the sternocleidomastoid
- Lateral to the carotid sheath, behind the submandibular gland
- Thick wall when infected

TOP DIFFERENTIALS

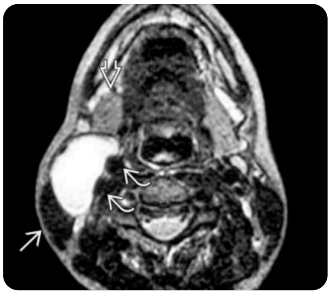
- Cystic nodal metastasis in an adult
- Lymphatic malformation
- Thymic cyst

CHECKLIST

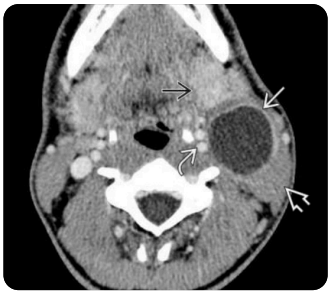
- In adults: exclude HPV oropharyngeal cancer
- Tract towards the tonsillar fossa



CT with intravenous contrast enhancement, coronal (a) and axial (b).
Dunnebier, Imaging for…



Axial T2WI MR shows a well-defined hyperintense mass anterior to the right SCM, posterolateral to the right submandibular gland, and…



Axial CECT in a 17-yearold boy shows a well-defined, low-attenuation mass in the typical location of a Bailey type II 2nd branchial cleft cyst: Anterior to t…



Coronal CECT in the same patient demonstrates a thicker wall inferiorly, which histologically represented lymphoid follicles in the wall of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 602.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

IMAGING FINDINGS

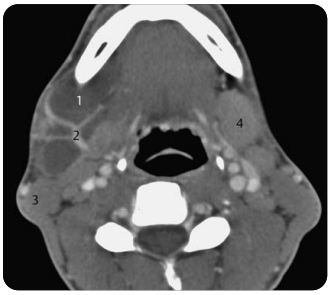
- Multicystic, septated, transspatial mass
- Fluid–fluid levels after bleeding
- Insinuates between structures without mass effect

TOP DIFFERENTIALS

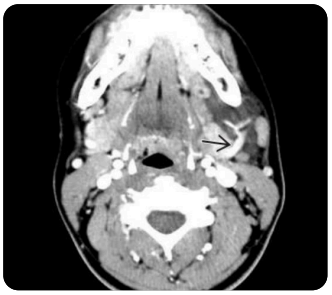
- Infantile haemangioma
- Venous malformation (phleboliths)
- Abscess

CHECKLIST

- Airway and mediastinal extent
- Macrocystic versus microcystic (sclerotherapy)



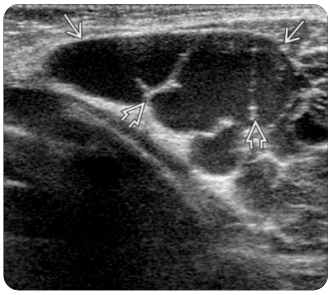
CT with intravenous contrast enhancement, axial. There is hypodense multilobular cystic structure (1) at the level of the…



Axial CECT in a 22-yearold woman with an angle of a mandible mass reveals a low-density submandibular space
Koch et al., Diagnostic Imaging: Head…



Axial T2 FS MR in the same patient shows the high fluid signal of the LM with its lobular margins. In this case, no fluidfluid levels are seen.…



Ultrasound in a 2-yearold patient with a soft, pliable mass shows a well-defined, lobulated, multicystic lesion with thin internal septations, typical o…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580.

LARYNX & TRACHEA · TUMOURS

Supraglottic Tumor

IMAGING FINDINGS

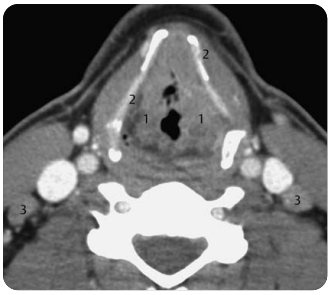
- Mass on the epiglottis, aryepiglottic folds or false cords
- Invasion of pre-epiglottic or paraglottic fat (T3)
- Nodes common, often bilateral

TOP DIFFERENTIALS

- Laryngocele
- Benign mucosal swelling
- Minor salivary tumour

CHECKLIST

- Vallecula and tongue base; pyriform sinus
- Cartilage and extralaryngeal spread



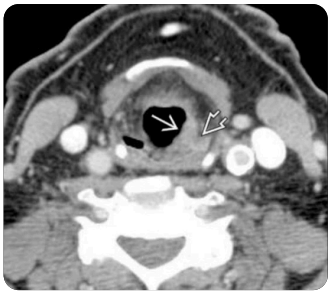
CT with intravenous contrast enhancement, axial. The lesion extends bilaterally (1), highly indicative of a malignancy. The…



Axial CECT shows similar findings with invasion of bilateral paraglottic fat and left AE fold. Tumor bulges through thyroid cartilage notch. Pyriform sinu…



Axial CECT shows small epiglottic mass, staged as T3 due to preepiglottic space invasion. Mass is symmetric, making it hard to…



Axial CECT shows small T1 posterior laryngeal SCCa with mass on left AE fold. Left pyriform sinus is collapsed, but no tumor is present in…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526.

LARYNX & TRACHEA · TUMOURS

Glottic Tumor

IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

CHECKLIST

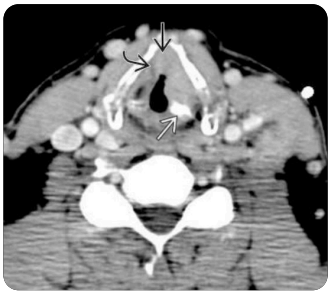
- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a



CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…



Axial CECT reveals SCCa involving the entire left true TVC, anterior commissure, and anterior 1/3 of right cord. Left arytenoid and thyroid…



Axial CECT shows bulky, ulcerated anterior SCCa involving both vocal cords and anterior commissure. Both anterior thyroid cartilages are sclerot…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.

For personal study. Follow local protocols and senior advice for patients.

THYROID & PARATHYROID · CONGENITAL

Thyroglossal Duct Cyst

IMAGING FINDINGS

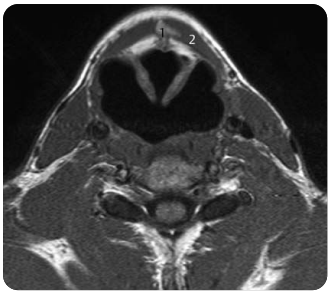
- Midline or paramedian cyst at or below the hyoid
- Infrahyoid cysts embedded in the strap muscles
- Thick wall or enhancement if infected

TOP DIFFERENTIALS

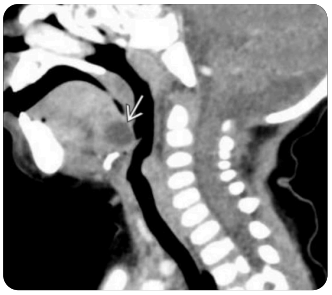
- Dermoid or epidermoid
- Lymphatic malformation
- Lingual thyroid

CHECKLIST

- Normal thyroid in the lower neck on ultrasound
- Solid nodule in the wall in an adult: carcinoma



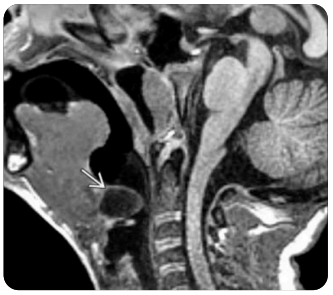
MRI, T1-weighted, axial. Small lesion (1) with high signal intensity on T1-weighted images, probably due to presence of protein-rich fluid, situate…



Sagittal CECT shows a well-defined cysticappearing mass at the midline base of the tongue. This was incidentally found on a CT performed…



Sagittal CECT in a child with sore throat and difficulty swallowing shows a lobulated low-attenuation mass in the midline sublingual space, with…



Sagittal T1WI in a child with recurrent TGDC demonstrates a cystic-appearing mass at the midline base of the tongue. Note distortion of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 590.

THYROID & PARATHYROID · INFECTION & INFLAMMATION

Hypertrophy of the Thyroid Gland, Thyroid Goiter

IMAGING FINDINGS

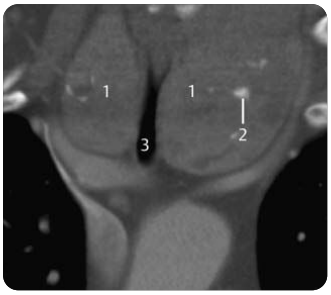
- Enlarged, heterogeneous gland with nodules, cysts and coarse calcification
- Tracheal deviation and narrowing
- Retrosternal extension

TOP DIFFERENTIALS

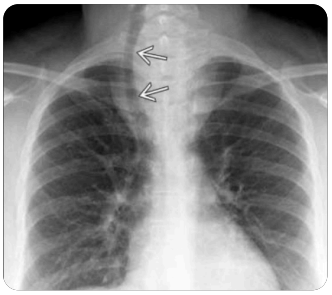
- Thyroid adenoma
- Differentiated or anaplastic carcinoma
- Lymphoma (rapid growth, Hashimoto)

CHECKLIST

- Lowest extent relative to the aortic arch and brachiocephalic vein
- Tracheal calibre for anaesthesia



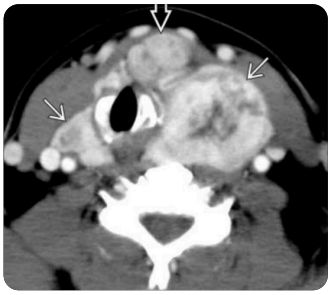
CT with intravenous contrast enhancement, coronal. This slice at the level of the lower neck/upper thoracic cavity, shows enlarged thyroi…



Posteroanterior chest radiograph demonstrates marked displacement of cervical and thoracic trachea by a large, left-sided neck & superior…



Coronal reformatted CECT in the same patient illustrates craniocaudad extent of the MNG with heterogeneous thyroid tissue bilaterally, but more…



Axial CECT at level of cricoid shows markedly heterogeneous, lobulated, and enlarged thyroid lobes as well as thyroid isthmus. No calcifications or…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 226.

LARYNX & TRACHEA · TRAUMA

Laryngeal Trauma

IMAGING FINDINGS

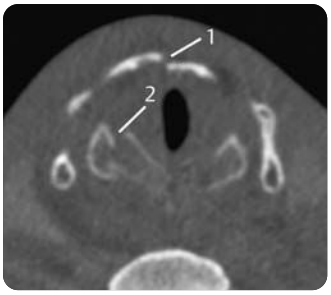
- Thyroid and cricoid cartilage fractures with displacement
- Arytenoid dislocation
- Haematoma and air narrowing the airway

TOP DIFFERENTIALS

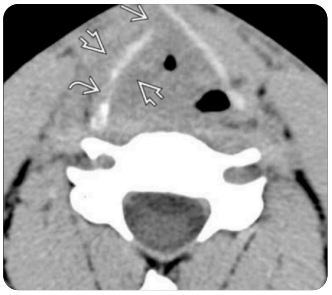
- Cord paralysis
- Unossified cartilage mimicking a fracture

CHECKLIST

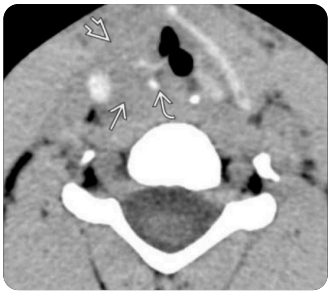
- Secure airway before scanning
- Cricoid fractures and displaced fragments usually need repair



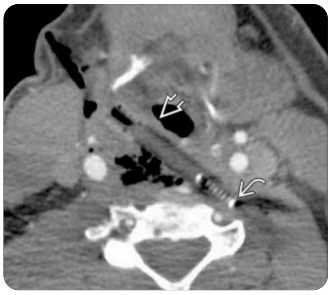
CT, axial bone window. CT at the level of the glottis showed multiple fractures; this figure shows a midline fracture with slight inward…



Axial NECT through the supraglottic larynx shows step-off at the superior thyroid notch. There is a subtle fracture of the right thyroid lamina…



Axial NECT more inferiorly in same patient, just above the cricoarytenoid joints, shows the anteromedial displacement of the right arytenoid…



Axial CTA performed for a pen stabbing through the supraglottic larynx shows displaced edematous right aryepiglottic fold. The pen tip…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialadenitis

IMAGING FINDINGS

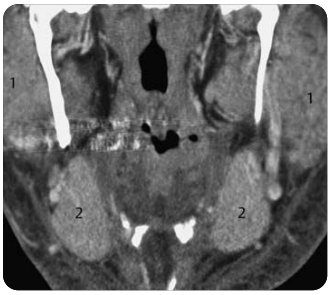
- Enlarged, enhancing gland with stranding of the surrounding fat
- Dilated duct, often with a stone
- Abscess: rim-enhancing collection

TOP DIFFERENTIALS

- Dental infection
- Nodal metastasis
- Gland tumour

CHECKLIST

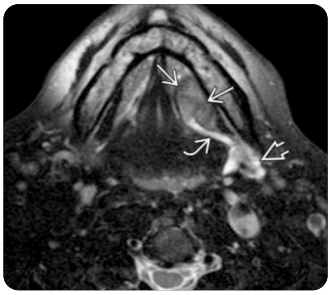
- Look for a stone along the duct
- Chronic: small, fibrotic gland



Patient with chronic intermittent enlargement of the salivary glands and diffuse pain.
Dunnebier, Imaging for…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…



Axial T2 FS MR demonstrates a well-circumscribed mass in the left floor of mouth in a patient with mucosal SCCa. The mass is obstructing the left…



Axial CECT through floor of mouth in a patient with right submandibular "mass" shows dense calculus in proximal duct at SMG hilus. Note…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialolith of the Submandibular Gland

IMAGING FINDINGS

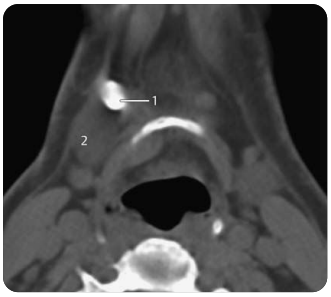
- Dense calculus in the Wharton duct or hilum on non-contrast CT
- Dilated duct proximal to the stone
- Enlarged inflamed or atrophic gland

TOP DIFFERENTIALS

- Phlebolith (venous malformation)
- Calcified node
- Dental fragment

CHECKLIST

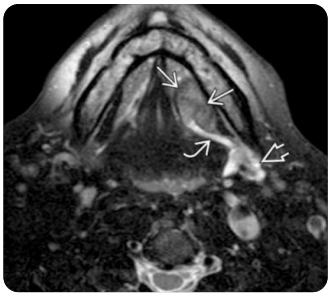
- Position relative to the mylohyoid and hilum
- Ultrasound shows stone and duct without radiation



CT, axial bone window.A hyperdensity (1), with similar density as bone, suggestive of a sialolith (stone, calculus), probably obstructin outflo…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…



Axial T2 FS MR demonstrates a well-circumscribed mass in the left floor of mouth in a patient with mucosal SCCa. The mass is obstructing the left…



Axial CECT through floor of mouth in a patient with right submandibular "mass" shows dense calculus in proximal duct at SMG hilus. Note…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Ranula of the Sublingual Gland

IMAGING FINDINGS

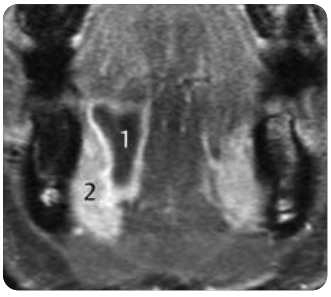
- Simple ranula: thin-walled cyst in the sublingual space
- Diving ranula: passes behind or through the mylohyoid into the submandibular space
- Tail sign pointing back to the sublingual space

TOP DIFFERENTIALS

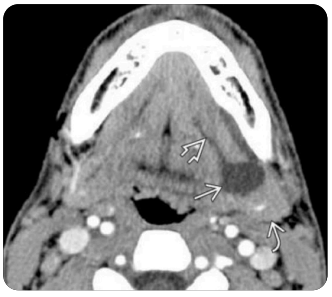
- Lymphatic malformation
- Dermoid or epidermoid
- Second branchial or thyroglossal cyst

CHECKLIST

- Relation to the mylohyoid decides the approach
- Remove the sublingual gland to prevent recurrence



MRI, T1-weighted with gadolinium enhancement and fat suppression, coronal. There is a cystic lesion suggestive of a ranula (1), originating…



Axial CECT demonstrates a rounded cystic mass in the submandibular space abutting the posterior margin of the mylohyoid muscle and displacing…



Axial CECT shows simple unilateral ranula as an ovoid, low-density lesion in left anterior SLS. There is no perceptible thickening or…



Axial CECT demonstrates a cystic, horseshoe-shaped lesion in anterior SLS, curving around and above genioglossus muscle insertion on…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400.

SALIVARY GLANDS · TUMOURS

Pleomorphic Adenoma of the Parotid Gland

IMAGING FINDINGS

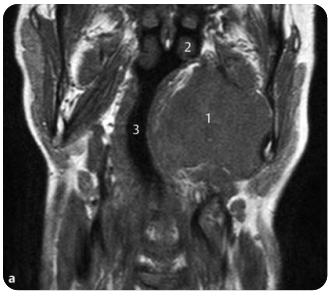
- Well-defined, lobulated mass with a capsule
- Very bright on T2
- Deep lobe: widens the stylomandibular gap, pushes parapharyngeal fat medially

TOP DIFFERENTIALS

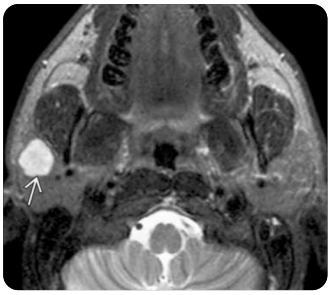
- Warthin tumour (tail, multiple, smokers)
- Low-grade mucoepidermoid carcinoma
- Intraparotid node

CHECKLIST

- Superficial or deep to the retromandibular vein
- Never enucleate: recurrence



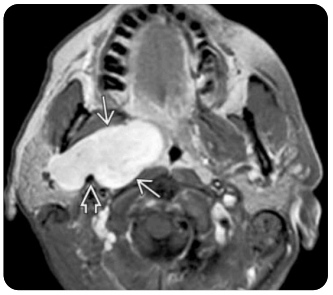
MRI, T1-weighted, coronal. Note the extensive, isointense, smooth-bordered lesion (1) in the hypopharyngeal area: from the level o...



Axial T2 FS MR reveals a sharply circumscribed, high-signal BMT in the superficial lobe of the parotid gland. This tumor at the very least abuts, if...



Axial T2 FS MR shows a pear-shaped mass extending from the deep lobe of the parotid gland, through the enlarged stylomandibular notch,...



Axial T1WI C+ FS MR shows a large, lobulated, homogeneously enhancing mass in the deep lobe of the parotid. The styloid process indents the...

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88.

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

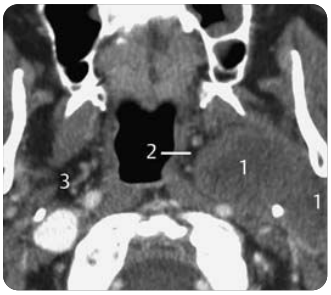
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

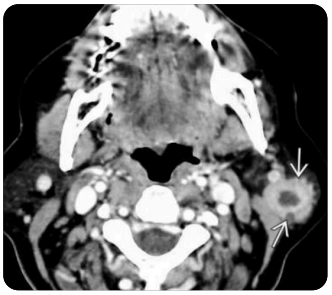
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

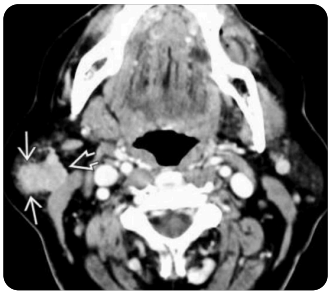
- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



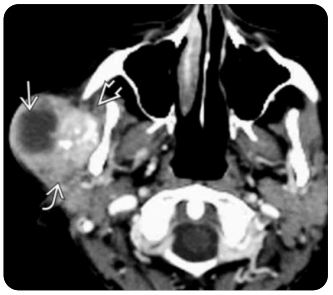
Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic ...



Axial CECT shows a mass in the parotid gland with ill-defined lateral borders and well-defined medial borders. The tumor enhances uniformly. Although...



Axial CECT shows a large, heterogeneous mass with cystic areas, heterogeneous enhancement, and ill-defined borders. The masseter...

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Zenker Diverticulum (Pharyngeal Pouch)

IMAGING FINDINGS

- Outpouching from the posterior hypopharynx above the cricopharyngeus on barium swallow
- Air–fluid level behind the oesophagus on CT
- Residual barium after the swallow

TOP DIFFERENTIALS

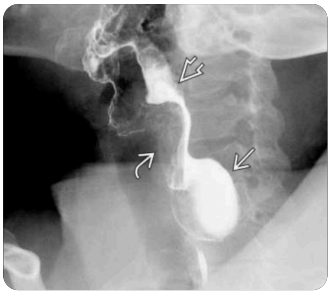
- Lateral pharyngeal or Killian–Jamieson diverticulum
- Paratracheal air cyst

CHECKLIST

- Size and neck of the pouch
- Aspiration on the swallow



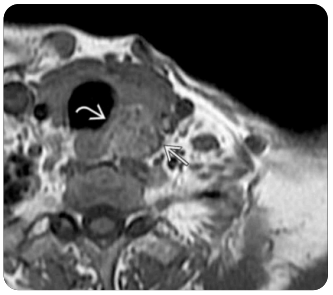
Barium swallow, anteroposterior view. After swallowing contrast, some contrast has remained in the hypopharynx (1) due to physiologic…



Barium esophagram demonstrates a large diverticulum with retained layering contrast from the posterior lateral junction of the hypopharynx…



Parasagittal T2WI MR shows a saccular well-circumscribed collection of heterogenous signal just below the cricoid cartilage at the level of C5-C6.…



Axial T1WI MR of brachial plexus in the same patient shows an incidental finding of a left Zenker diverticulum. Image demonstrates the mouth of th…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 258.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Laryngocele

IMAGING FINDINGS

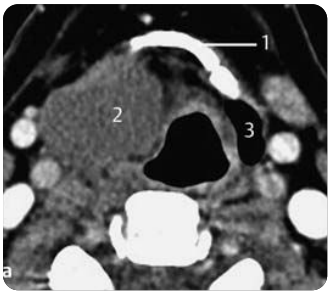
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

TOP DIFFERENTIALS

- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise



CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…

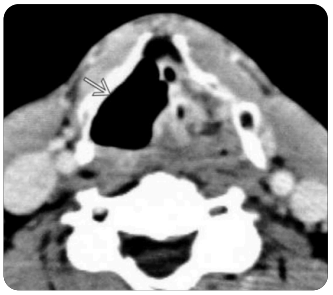


Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…



Axial CECT demonstrates the typical findings of a fluid-filled, internal

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 292



Axial CECT in a professional trumpet player shows an airfilled sac confined to the right paraglottic region. Findings are typical of an air-filled, internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Paralysis of the Recurrent Laryngeal Nerve

IMAGING FINDINGS

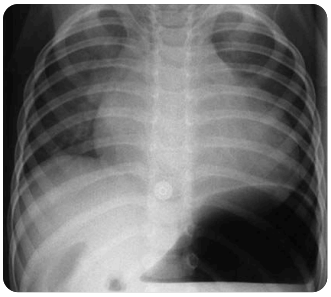
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



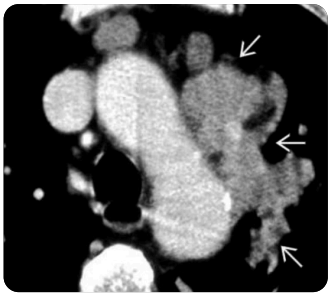
A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…



Axial CECT in the same patient demonstrates asymmetrically enlarged left pyriform sinus and a medial position of the left aryepiglotti…



Axial CECT in same patient stresses the importance of scanning the aortopulmonary window for left VCP. A mediastinal mass…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.

PHARYNX, TONSILS & OESOPHAGUS · TRAUMA

Foreign Bodies

IMAGING FINDINGS

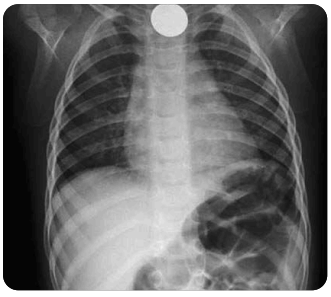
- Radio-opaque body on the lateral neck film; fish bones often faint
- CT shows the object, its position and depth
- Prevertebral swelling, air or abscess after perforation

TOP DIFFERENTIALS

- Calcified cartilage (thyroid, cricoid, arytenoid)
- Calcified stylohyoid ligament

CHECKLIST

- Low-dose CT if the film is negative and suspicion is high
- Perforation: mediastinal air, collection



Conventional radiography can still be convenient for evaluation of radiopaque foreign bodies, as in this 2-year-old child, who was suspected of…

PARANASAL SINUSES · INFECTION & INFLAMMATION

Sinusitis and its orbital and intracranial complications

IMAGING FINDINGS

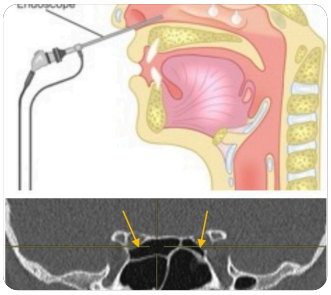
- Orbit: preseptal or postseptal stranding, subperiosteal or orbital abscess
- Intracranial: epidural or subdural empyema, abscess, meningitis
- Venous: superior ophthalmic vein and cavernous sinus thrombosis

TOP DIFFERENTIALS

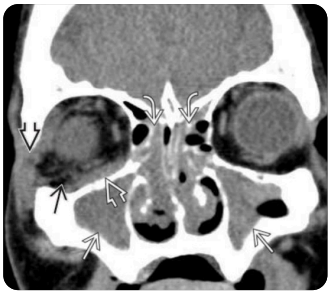
- Idiopathic orbital inflammation
- Tolosa-Hunt syndrome

CHECKLIST

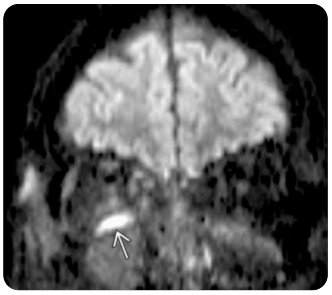
- Contrast CT for the orbit, MRI for intracranial disease
- Frontal sinusitis: look for Pott puffy tumour



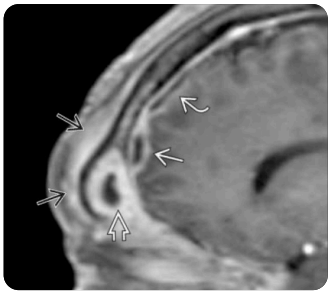
Coronal reconstructed image from a high-resolution CT. Bilateral Onodi cells (arrows). This puts the optic nerves at risk of damage during FESS…



Coronal NECT in a man with acute sinusitis complicated by orbital cellulitis and subperiosteal abscess shows extensive opacification of…



Coronal DTI trace in same patient shows reduced diffusivity, confirming subperiosteal abscess complicating acute sinusitis.…



Sagittal SPGR C+ in frontal sinusitis complicated by subdural empyema, meningitis, cerebritis, and subgaleal abscess shows frontal sinus mucosal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 696.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Otomastoiditis and its complications

IMAGING FINDINGS

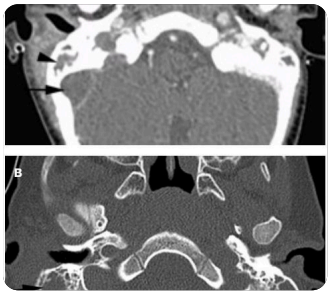
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

CHECKLIST

- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



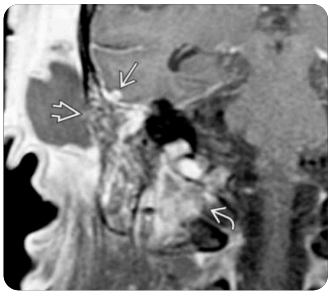
Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…



Axial bone CT of the same patient shows loss of mastoid trabecula and a dehiscent sigmoid plate diagnostic of coalescent otomastoiditis. Subtle…



Coronal T1WI C+ FS MR reveals mastoid enhancement with a large periauricular abscess. Lateral mastoid is focally dehiscent with thick and enhancing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

ORAL CAVITY & TONGUE · AFTER SURGERY

Squamous cell carcinoma: oral cavity, oropharynx and larynx

IMAGING FINDINGS

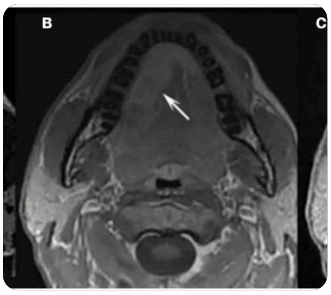
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

CHECKLIST

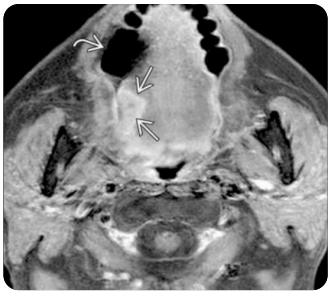
- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT



Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …



Axial T1WI C+ FS MR in the same patient shows marked enhancement a of wedge-shaped ulcer at posterolateral tongue surface. No…



Axial T2WI FS MR demonstrates a small right lateral tongue tumor, subtly evident as lenticular region of increased signal intensity. Ill-defined…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.

PARANASAL SINUSES · TUMOURS

Sinonasal tumours

IMAGING FINDINGS

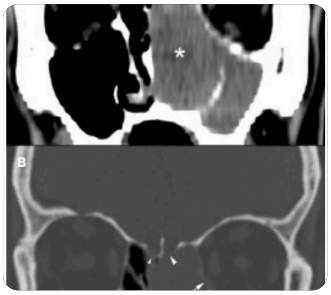
- Unilateral solid mass with bone destruction
- Intermediate T2, solid enhancement, distinct from bright secretions
- Spread to orbit, skull base, pterygopalatine fossa, cheek

TOP DIFFERENTIALS

- Inverted papilloma
- Lymphoma
- Olfactory neuroblastoma, adenocarcinoma, melanoma

CHECKLIST

- Periorbita and dura
- Perineural spread along V2 and the vidian nerve; nodes



Coronal reformatted images from a noncontrast CT in soft tissue window (a) and bone window settings (b). The mass (asterisk) causes resorption of…



Axial CECT shows the typical location and appearance of an antral SCCa. There is extension into the premaxillary soft tissues anteriorly an…



Axial T1 MR shows a large antral SCCa. The signal of the mass is similar to other soft tissues. There is extension anteriorly into the premaxillary soft…



Coronal T2 FS MR of SCCa demonstrates ethmoid sinus involvement and masticator space extension. The low T2 signal of this…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 754.

THYROID & PARATHYROID · TUMOURS

Thyroid cancer

IMAGING FINDINGS

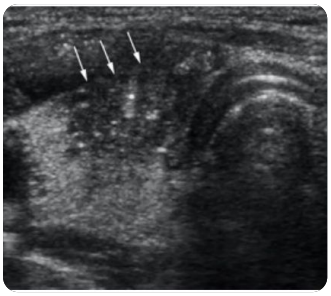
- Solid hypoechoic nodule with microcalcifications, taller than wide
- Irregular margins or extrathyroidal extension
- Cystic or calcified nodes in levels III, IV and VI

TOP DIFFERENTIALS

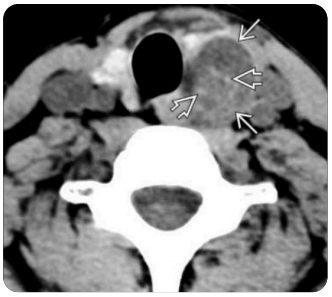
- Benign adenoma or colloid nodule
- Multinodular goitre
- Medullary or anaplastic carcinoma

CHECKLIST

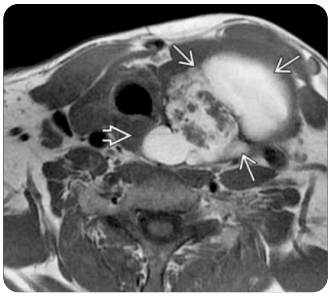
- Trachea, oesophagus, larynx and recurrent nerve
- Iodinated contrast delays radioactive iodine



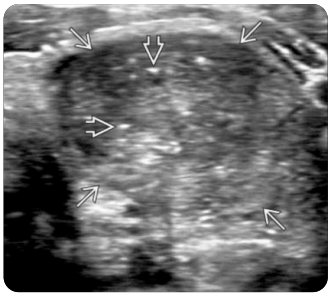
16-year-old lady presenting with hard right neck lump confirmed to be papillary ca on cytology. US shows strongly hypoechoic lesion with…



Axial NECT shows a well-defined mass arising from the left thyroid lobe with fine, speckled microcalcifications centrally. Small calcifications such as…



Axial T1 MR in a 49-yearold man with an enlarging neck mass shows a heterogeneous solid and cystic mass arising in left thyroid lobe, displacing…



Longitudinal thyroid ultrasound in a different patient reveals a well-defined mass with multiple tiny, hyperechoic microcalcifications, found to be a…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238.

IMAGING

Advanced and rare

Once the core is secure. Each has its full set of images in the app.

Inclusion Cholesteatoma and Atresia of the External Auditory Canal

External ear & tympanic membrane · Congenital

Expansion toward and/or destruction of the temporomandibular joint and toward the middle ear structures, abscess formation, and osteomyelitis and (intracranial) spread of infection. · In case of aural atresia (first branchial groove anomaly), other dysmorphic features may be present too, especially in case of syndromal comorbidity.

Exostoses of the External Auditory Canal

External ear & tympanic membrane · Bone disease

In patients with aural discharge, chronic otitis may be the result of infected epithelial stasis. · In cases with a narrowed orifice and meatus, a meatoplasty might be considered for better aeration and cleaning options. Furthermore, CT might be reassuring, showing a normal aerated middle ear.

Stenosis of the External Auditory Canal

External ear & tympanic membrane · Bone disease

Beware of accumulation of cerumen or epithelium behind the fibrous wall, and exclude deeper pathology in the middle ear. In most cases, the fibrous wall can be surgically separated from the inner layers of the eardrum without opening the middle ear. · In cases of malignant external otitis or secondary cholesteatoma, deeper pathologies (e.g., to the facial nerve) are of special interest.

Malignancy of the External Auditory Canal Squamous Cell Carcinoma

External ear & tympanic membrane · Tumours

Extension of bony destruction and features suggestive of infiltrative growth are associated with high morbidity, such as dysfunction of the facial nerve and inner ear. · Adherence to vital structures, such as the internal carotid artery, may preclude surgery.

Fixation of the Ossicular Chain

Middle ear & mastoid · Bone disease

As mentioned above in differential diagnosis, particular attention must be paid to patients with conductive hearing loss in combination with a normal appearing eardrum and aerated middle ear. · In case of ankylosis of the ossicular chain, accurate evaluation of thin axial and coronal slices may reveal deformities and/or fixation.

Stapedectomy, Control of Piston Position

Middle ear & mastoid · After surgery

Ossicular prostheses may differ as regards their degree of radiopacity and visibility on CT. For this reason fine slices are required. · Malpositioning of the piston around the incus and possible erosions or resorbtion of the long process and lenticular process due to compression of the piston are difficult to evaluate on CT.

Osteogenesis Imperfecta

Middle ear & mastoid · Bone disease

In osteogenesis imperfecta affecting the middle ear, fractures of the stapedial crus are often found on middle ear inspection and/or fixation of the footplate in combination with a known (family) history of osteogenesis imperfecta. Note the typical blue sclerae of the eyes in osteogenesis imperfecta type I. · Extension of demineralization, which may be unilateral or bilateral and may vary between both ears. Most cases of osteogenesis imperfecta have minor abnormalities as described in Pathology of the Middle Ear earlier in this chapter. A clinical history of frequent and multiple fractures due to minor trauma everywhere in the body, as well as the typical blue sclerae, might help make the diagnosis.

Gusher

Inner ear & IAC · Infection & inflammation

A wide communication between the subarachnoid space and the perilymph (a defect at the fundus of the IAC or a dilated cochlear aqueduct) lets CSF pour out when the footplate is opened. · CT clues: a bulbous, widened lateral IAC with a deficient fundus, absent modiolus and a corkscrew cochlea (X-linked incomplete partition type III, POU3F4).

Persistent Stapedial Artery

Middle ear & mastoid · Variants

An embryonic branch of the ICA (from the hyoid artery) that persists, crossing the promontory and passing through the obturator foramen of the stapes. · CT: a small canal over the promontory, an enlarged anterior tympanic segment of the facial canal, and an absent foramen spinosum (the middle meningeal artery arises from the stapedial artery instead).

Adenoma of the Middle Ear

Middle ear & mastoid · Tumours

Evaluation of extension to critical structures such as the internal carotid artery and facial nerve is essential when considering whether complete surgical removal is possible.

Malformations, Treacher Collins

Middle ear & mastoid · Congenital

In deformities of the temporal bone, pay particular attention to the: · External auditory canal, bony or fibrous atresia, appearance and position of the mandibular condyle and temporomandibular joint.

Fibrous Dysplasia of the Temporal Bone and Skull Base

Temporal bone & skull base · Bone disease

Radiologically on CT, enlargement of bone with intact cortical outlines and a dense ground glass appearance is characteristic of fibrous dysplasia. Fibrous dysplasia incidentally found on MRI may be confused with a malignancy. · In meningioma, bony outlines can be irregular and signs of hyperostosis can be found as well as a dural tail. Sometimes intracranial calcifications might be indicative of an intracranial meningioma.

Iatrogenic Fenestration

Middle ear & mastoid · After surgery

In fenestrated patients, no actual compression due to masses will be seen. · Beware of suction of the cavity for cleaning. This must be performed very carefully to prevent severe vertigo and inner ear dysfunction.

Modified Radical Cavity, Persistent Infections

Middle ear & mastoid · After surgery

Attention must be paid to the above-mentioned presence of residual cells. Residual cholesteatoma may present as smooth-bordered, rounded swellings covered by an epithelial or mucosal lining. In cases of destruction of the bony outline to the middle of posterior cranial fossa, a complementary MRI might be helpful for evaluating the extent of the pathology and its relation to intracranial structures. · DWI is especially helpful in discriminating between recurrent or residual cholesteatoma and granulation tissue.

Osteoradionecrosis

Middle ear & mastoid · Bone disease

Stasis of secretions and infections are frequently the result of eustachian tube dysfunction due to mucosal swelling and loss of ciliary function. Chronic infections may require complete surgical removal of the affected bone and long-term antibiotic treatment.

Dural Prolapse and Cerebrospinal Fluid Leakage

Middle ear & mastoid · After surgery

Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement.

Hemangioma of the Geniculate Ganglion

Facial nerve · Tumours

Signs of infiltration or expansion. An MRI may be necessary to exclude or evaluate intracranial involvement. · Most tumors, such as schwannoma, demonstrate more sharply defined enhancement.

Schwannoma of the Facial Nerve

Facial nerve · Tumours

CT and MRI are complementary in the evaluation of the extension of the schwannoma to the middle or posterior cranial fossa as well as to the inner ear structures. Compared with hemangiomas, schwannomas of the facial nerve can be quite large before symptoms occur. · Surgical removal with facial nerve grafting is only considered in (near) total paralysis of the facial nerve or when expansion of the lesion would lead to the damage of surrounding critical structures.

Endolymphatic Sac Tumour

Inner ear & IAC · Tumours

Signs of bone destruction in the region of the jugular bulb and jugular foramen. – · Cranial nerve VII XI palsies, extension into structures of the neck and skull base.

Malformations of the Inner Ear and Ossicular Chain

Inner ear & IAC · Congenital

This section illustrates the importance of a systematic comparison of both sides as well as the different compartments of the ear since bilateral congenital pathology may show (subtle) variations between the left and right side.

Internal Auditory Canal, Enlargements

Inner ear & IAC · Congenital

An internal auditory canal width of less than 3 mm is considered pathologic and may also indicate hypoplasia or absence of its contents. · A width of more than 10 mm may also be pathologic, and assessment of the bony outline of the meatus is essential in the evaluation of pathology. Intralesional calcifications and hyperostosis are suggestive of meningioma.

Otosclerosis, Retrofenestral

Inner ear & IAC · Bone disease

Extension of demineralization. It may be unilateral or bilateral and may vary between both ears. Most cases of otosclerosis demonstrate minor deformities as described in Pathology of the Middle Ear earlier in this chapter. · In cases of severe bilateral hearing disorders or complete deafness, cochlear implantation may be considered, although the patency of the cochlear lumen is essential for correct and complete electrode insertion. There is a risk of going down a false route due to the severe demineralization and distortion of the normal anatomy.

Cochlear Implantation

Inner ear & IAC · After surgery

Schwannoma of the Internal Auditory Canal

Inner ear & IAC · Tumours

Vestibular schwannoma is the most frequent pathology (90%) found in this area. The clinical presentation facilitates the diagnosis because of hearing problems (vestibulocochlear). · Multiple cranial nerve deficits are suggestive of inflammatory processes.

Cochlear Schwannoma

Inner ear & IAC · Tumours

Subtle and diffuse enhancements are also suggestive of active immunologic pathologies or intralabyrinthine infections. · Fibrosis or ossification may occur over a period of months, excluding the possibility of cochlear implantation in bilateral deafness.

Schwannoma of the Cerebellopontine Angle

Inner ear & IAC · Tumours

A vestibular schwannoma is the most frequent pathology (90 %) in this area. Its initial clinical presentation will help to determine the origin of the lesion, that is of hearing problems (vestibulocochlear) or facial paresis. · Meningioma may show the so-called dural tail, although this is not always present, and involvement of the internal auditory canal is less common.

Arachnoid Cysts

Inner ear & IAC · Infection & inflammation

Epidermoid cysts, which are also hypointense on T1-weighted sequences and hyperintense on T2-weighted sequences, can be differentiated with diffusion-weighted imaging (DWI). · In contrast, lipomas are hyperintense on T1-weighted images and hypointense on spin-echo T2-weighted sequences.

For personal study. Follow local protocols and senior advice for patients.

Cholesteatoma with Intracranial Involvement

Middle ear & mastoid · Infection & inflammation

Epidermoid cysts and cholesterol granulomas are more frequently located in the petrous apex. · Endolymphatic sac tumor originates in the region between the posterior semicircular canal and the middle cranial fossa and sigmoid sinus, with typical imaging patterns (see separate sections).

Cystic Degeneration in CPA Schwannoma

Inner ear & IAC · Tumours

An arachnoid cyst may be secondary to folding or inclusion of arachnoid membranes by compression from a schwannoma or meningioma. They are mostly located outside the margin of the causative lesion. · A mucocele may be considered when the cyst is in located in a structure covered by mucosa.

Effects of Radiotherapy on CPA Schwannoma

Inner ear & IAC · After surgery

After stereotactic radiosurgery or fractionated radiotherapy, schwannomas often swell transiently for 6–18 months (pseudoprogression) and lose central enhancement before shrinking. · Do not call failure on early growth alone; follow with serial MRI, and consider treatment failure only with sustained growth over about 2–3 years.

Neurofibromatosis II (NF II)

Inner ear & IAC · Tumours

Neurofibromatosis may initially present as otologic deficits. – · The only ENT finding in cases of von Hippel–Lindau disease is the presence of a unilateral or bilateral endolymphatic sac tumor.

Trigeminal Nerve Schwannoma

Trigeminal nerve · Tumours

Initial and later clinical symptoms are suggestive of the site of origin of the lesion as well as helping in differentiation between a tumor and inflammatory lesion. · Multiple lesions or cranial nerve deficits and rapid progression are indicative of inflammatory disorders.

Astrocytoma in the Cerebellopontine Angle

Inner ear & IAC · Tumours

Sharpness of the margins helps to differentiate between a benign and malignant lesion. · An endolymphatic sac tumor originating near the osseous mastoid part behind the labyrinth is mostly confined extradurally.

Labyrinthitis and Subclinical Meningitis

Inner ear & IAC · Infection & inflammation

Otoscopy may exclude middle ear infections although CT may be indicated in some cases. MRI will show (retro)cochlear enhancement in cases of (acute) infections. · Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and ossification. In bilateral cases, this may hamper insertion of a cochlear implant.

Otosyphilis and Intracranial Complications

Inner ear & IAC · Infection & inflammation

Cranial nerve deficits, rapid progression, and multiple lesions are strongly indicative of inflammatory disorders. If malignancy is suspected, beware of involvement of other organs as primary sites of possible metastases. · Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and/or ossification. In bilateral cases, this may hamper insertion of a cochlear implant, although its success may also be restricted by retrocochlear deficits.

Infection of the Skull Base and Gradenigo Syndrome

Temporal bone & skull base · Infection & inflammation

In Gradenigo syndrome, the dural branches of the first division of the trigeminal nerve are involved in the symptoms of pain; the abducens nerve is paralyzed because of compression in the Dorello canal in the petrous apex. Pseudo-Gradenigo may be due to a nasopharyngeal carcinoma infiltrating the skull base. · Malignant otitis externa may lead to skull-base infection. Immunocompromised and diabetic patients are prone to bacterial (e.g., Pseudomonas) or fungal (e.g.,) infections. Sequential radiographic and radionuclide (e. Aspergillus g., technetium and gallium) imaging may be used to monitor the efficacy of treatment. Complications may result from the spread of infection to adjacent intracranial or head and neck structures.

Petrositis

Temporal bone & skull base · Infection & inflammation

Clinical symptoms may help differentiate between inflammatory disorders, asymptomatic pseudotumors, and other pathology. CT and MRI are complementary. Knowing whether the origin is intradural or extradural and the presence of bony changes help to establish the most likely diagnosis.

Meningioma with Skull Base Destruction

Paranasal sinuses · Tumours

On CT, the bone changes associated with hyperostosing meningioma en plaque may be confused with (polyostotic) fibrous dysplasia. However, in meningioma the cortical margins of the affected bones are not well delineated. · In fibrous dysplasia, the cortical margins are preserved (compare with section on fibrous dysplasia). Cranial nerves and intracranial vasculature may be encroached on by the lesion. Total surgical removal is rarely possible.

Mucocele of the Frontal Recess

Paranasal sinuses · Infection & inflammation

Beware of large osseous lesions of the skull base, removal of which may result in protrusion of intracranial contents or marsupialization of the mucocele. · In meningo(encephalo)cele or intracranial tumors, persistent leakage of cerebrospinal fluid may occur at biopsy.

Olfactory Neuroblastoma (Olfactory neuroblastoma (esthesioneuroblastoma))

Nose & nasal cavity · Tumours

Olfactory neuroblastoma (esthesioneuroblastoma) is an uncommon tumor of neural crest origin that arises from the olfactory mucosa in the superior nasal cavity. These polypoid tumors typically start unilaterally and may bleed profusely on biopsy. Neglected cases appear as a bilateral mass in the anterior cranial fossa. In these larger tumors, which usually have intracranial extension, peripheral tumoral cysts can occur at the margins of the intracranial part of the mass. These cysts have their broadest base on the tumor and, when present, are highly suggestive of the (imaging) diagnosis of olfactory neuroblastoma. Beware of encroachment of cranial nerves or intracranial vessels, which may be important in the estimation of morbidity during removal.

Pituitary Gland Pathology

Paranasal sinuses · Tumours

In classifications of pituitary gland pathology, lesions within the sella are classified as microadenomas. Microadenomas are frequently active hormone-producing tumors with specific clinical symptoms (e.g., galactorrhea). · Tumors extending out of the sella are classified as macroadenomas. Most macroadenomas are nonfunctional tumors. Therefore, these lesions present late and can reach remarkable volumes. Large pituitary adenomas may be accompanied by cystic degeneration with or without hemorrhage. Macroadenomas with superior extension may compromise the optic chiasm with visual field disturbances and even blindness. Less frequently, macroadenomas spread inferiorly into the sphenoid sinus.

Intracranial Abscess from an Otologic Focus

Middle ear & mastoid · Infection & inflammation

Otogenic abscesses lie in the temporal lobe (through the tegmen) or the cerebellum (through Trautmann's triangle or the sigmoid sinus). · MRI: a ring-enhancing lesion with restricted diffusion in the centre (unlike necrotic tumour) and surrounding oedema.

Intracranial Abscess from a Sinonasal Focus

Paranasal sinuses · Infection & inflammation

Frontal and ethmoid sinusitis spread through valveless diploic veins or bone erosion to cause epidural, subdural or frontal lobe abscess (Pott's puffy tumour is the extracranial sign). · Contrast MRI is more sensitive than CT for subdural empyema and cerebritis; repeat imaging if seizures or reduced consciousness develop after sinus surgery.

Intracranial Hemorrhage Due to a Via Falsa

Temporal bone & skull base · After surgery

During mastoidectomy, drilling through the tegmen or the sinodural angle without recognising the middle fossa dura can enter the temporal lobe ('via falsa'). · Find the tegmen early, thin it under irrigation and follow it, keeping the dura in view; a low-lying tegmen is visible on preoperative coronal CT.

Intracranial Hemorrhage due to a Middle Cranial Fossa Approach

Temporal bone & skull base · After surgery

The middle fossa approach needs temporal lobe retraction, so postoperative imaging may show contusion, extradural or intracerebral haematoma. · CSF leakage through a tegmen defect into a radical cavity shows as fluid in the cavity with a bony gap on coronal CT; MRI or cisternography confirms the dural defect.

Dural Defect after Sinonasal Surgery

Paranasal sinuses · After surgery

The commonest site of iatrogenic skull base injury is the lateral lamella of the cribriform plate (thinnest bone, where the anterior ethmoidal artery enters), then the posterior ethmoid roof. · Locate the defect with thin-section coronal CT; MR cisternography (heavily T2-weighted) shows CSF tracking through it; beta-2 transferrin confirms leaked fluid.

Fracture of the Nasal Bone and Retention Cysts

Nose & nasal cavity · Trauma

In cases of large cysts resulting in obstruction of the maxillary sinus and those accompanied by signs of sinusitis, surgical removal (marsupialization) might be considered to improve drainage.

Unilateral Nasal Polyps

Nose & nasal cavity · Infection & inflammation

Destruction of bony outlines (with infiltrative characteristics) is suggestive of inverted papilloma, malignancies, or fungal infection. · Calcifications in the opacified area are suggestive of fungal infection. Resorption of bone around tooth roots suggests odontogenic infection.

Frontal Sinusitis

Paranasal sinuses · Infection & inflammation

The anatomy and pathology of the frontal recess required special attention. · In cases of persisting drainage problems in the region of the frontal recess, the septum between the frontal sinuses may be removed for establishing drainage by way of the left frontal recess.

Odontogenic Sinusitis

Paranasal sinuses · Infection & inflammation

Destruction of bony outlines with infiltrative characteristics is suggestive of inverted papilloma, malignancies, or fungal infections. · Calcifications in the opacified area are suggestive of fungal infections.

Cystic Fibrosis

Paranasal sinuses · Infection & inflammation

Destruction of bony outlines with infiltrative signs is more characteristic of malignancy. · Expansion is more likely to be due to benign processes, such as polyposis or mucocele.

Granulomatosis with Polyangiitis (Wegener)

Nose & nasal cavity · Infection & inflammation

A history of chronic sinusitis in combination with destruction of the nasal septum is highly suggestive of granulomatosis with polyangiitis (GPA). Pulmonary complaints and rapid improvement after steroid treatment may confirm this diagnosis. These therapeutic effects may also be observed in cases of allergic fungal sinusitis. · Orbital involvement appears as a mass resulting in protrusion of the eyeball and diplopia (see also granulomatosis with polyangiitis (GPA), Orbital Involvement).

Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)

Paranasal sinuses · Bone disease

· Expansion of bone with a ground glass appearance and intact cortex reflects the pathophysiology of this disease (i.e., replacement of normal medullary bone by woven fibro-osseous tissue).

Osteoma

Paranasal sinuses · Bone disease

Large osteomas may cause functional problems due to growth and encroachment of surrounding structures (see Osteoma, Gardner Syndrome, and Complications). For this reason, and also in small osteomas, a wait-and-scan policy might be advisable.

Piriform Aperture Malformation

Nose & nasal cavity · Congenital

A developmental disorder of this kind may result in concomitant pathologies, such as the fusion of both maxillary central incisors to form one megaincisor (Fig. 7.15), in the majority of patients. Holoprosencephaly and abnormalities of the pituitary–adrenal axis are also encountered.

Nasolacrimal Mucocele

Orbit & sinus relations · Congenital

Thornwaldt Cyst

Pharynx, tonsils & oesophagus · Congenital

Myxoma

Paranasal sinuses · Tumours

Osteoma, Gardner Syndrome, and Complications

Orbit & sinus relations · Bone disease

Extension to structures which, at surgical removal, may result in cosmetic or functional complications. · In Gardner syndrome, multiple osteomas may be found as well as other pathologies, some of which are at an increased risk of malignancy (colorectal carcinoma and thyroid carcinoma).

Granulomatosis with Polyangiitis, Orbital Involvement

Nose & nasal cavity · Infection & inflammation

Amyloidosis in the Orbit

Orbit & sinus relations · Infection & inflammation

Differentiation between solitary involvement of the orbit and involvement or growth from adjacent structures. · Differentiation between unilateral and bilateral disease.

Anterior Skull Fractures

Orbit & sinus relations · Trauma

Encroachment of functional structures (ocular muscles, cranial nerves) or compression due to hematoma (orbit, subdural). · Displacements resulting in functional (movements of the eyeball, obstruction of the nose) or cosmetic deficits.

For personal study. Follow local protocols and senior advice for patients.

Orbital Decompression in Graves Disease

Orbit & sinus relations · After surgery

Conventional sinonasal radiography in these patients is not sufficient for the evaluation of concomitant sinusitis. CT will be needed for anatomic orientation, especially if endoscopic surgery is contemplated. · Endonasal surgical procedures must be done with care due to distortion of the normal anatomy and an increased risk of damage to displaced intraorbital structures such as the optic nerve.

Malignancy of the Pterygopalatine and Infratemporal Fossa

Infratemporal fossa & TMJ · Tumours

Multiple Synchronous Malignancies

Pharynx, tonsils & oesophagus · Tumours

Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and spread of each tumor to the contralateral side. · Lymphomas (non-Hodgkin) initially may present as unilateral lymphadenopathy and ulcerative lesions.

Lymphadenopathy, HIV

Neck: triangles, fascia & nodes · Infection & inflammation

HIV infections may be associated with coexisting lymphoma and chronic lymphadenitis. Coexisting diseases may have a different clinical presentation in HIV.

Supraglottic and Pharyngeal Tumors

Larynx & trachea · Tumours

Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and overgrowth of the tumor on the contralateral side. · Lymphomas (non-Hodgkin) initially might present as unilateral lymphadenopathy and ulcerative lesions.

Pleomorphic Adenoma of the Submandibular Gland

Salivary glands · Tumours

Calcifications are well depicted on CT, as are signs of infection and abscess formation. On CT, most pleomorphic adenomas are smooth-bordered, spherical, benign-looking tumors that usually display a higher density than the surrounding parotid parenchyma. Sometimes, these tumors demonstrate a heterogeneous appearance with sites of lower attenuation representing areas of necrosis, old hemorrhage, and/or cystic change (see · For MRI characteristics, see Pleomorphic Adenoma of the Parotid Gland, page 316.

Lipoma Near the Parotid Gland

Salivary glands · Tumours

Lipomas have fat density on CT (−50 to −100 HU) and follow fat signal on every MRI sequence, with suppression on fat-saturated images. · Thin septa are acceptable; thick enhancing septa or nodules suggest liposarcoma.

Cholesteatoma Near the Parotid Gland

Salivary glands · Infection & inflammation

A cholesteatoma or epidermoid can extend from the ear canal or mastoid into the parotid region, or arise from a first branchial cleft remnant. · CT: a smooth, expansile low-density lesion with bone erosion of the canal or tympanic plate; MRI: restricted diffusion on DWI, the key feature.

Adenoid Cystic Carcinoma

Salivary glands · Tumours

Slow-growing but relentless, with perineural spread along VII (to the stylomastoid foramen and geniculate ganglion) and V3 (foramen ovale) as its hallmark; skip lesions are common. · MRI with fat-saturated post-contrast T1 shows nerve thickening and enhancement and widening of the foramina; image the whole nerve to the brainstem.

Acinic Cell Carcinoma

Salivary glands · Tumours

A low-grade malignancy, mostly in the parotid, often well circumscribed and so easily mistaken for a benign tumour on imaging. · May be cystic or multifocal; up to a few per cent are bilateral.

Aspiration

Larynx & trachea · Infection & inflammation

Occasionally aspiration is caused by mechanical failure of the epiglottis to close because of the presence of tumor, but most cases are due to neurologic (sensory) disorders without any visible pathology apart from the aspiration. Beware of aspiration pneumonia. · The above-mentioned pathologies are partly demonstrated in other sections.

Esophageal Passage Disorders

Pharynx, tonsils & oesophagus · Infection & inflammation

Patients with removable dentures may lack oral sensory input regarding the size of their food. This may also happen with sharp foreign bodies, which may cause perforations of the esophagus. · Anatomic structures such as the aortic arch or an aberrant right esophageal artery are demonstrated as persistent impressions without stasis of contrast.

Pneumothorax after Neck Surgery

Neck: triangles, fascia & nodes · Trauma

Most cases of pneumothorax are much smaller than in the picture shown here, and most of them are located at the lung apex.

Lipoma of the neck

Neck: triangles, fascia & nodes · Tumours

Schwannomas of the oral cavity and neck

Neck: triangles, fascia & nodes · After surgery